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Ordinal Regression Heads and Loss Functions for Fine-Grained Arabic Readability Assessment

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Dedication

Dedication

Acknowledgment

List of abbreviations and acronyms

AI	*Artificial Intelligence*
DL	*Deep Learning*
ML	*Machine Learning*
MSA	*Modern Standard Arabic*
NLP	*Natural language processing*
PLM	*Pretrained Language Model*
BERT	*Bidirectional Encoder Representations from Transformers*
AraBERT	*Arabic BERT*
BAREC	*Balanced Arabic Readability Evaluation Corpus*
QWK	*Quadratic Weighted Kappa*
CE	Cross Entropy
MSE	Mean Squared Error
EMD	Earth Mover's Distance

OLL	Ordinal Log Loss
CORAL	Consistent Rank Logits, a rank-consistent ordinal regression head
D3Tok	Morphologically Segmented Tokenization
IAA	Inter-Annotator Agreement
GPU	Graphics Processing Unit
CPU	Central Processing Unit
POS	Part of Speech
OOV	Out Of Vocabulary
Soft-QWK	*Differentiable Soft Quadratic Weighted Kappa Loss*
SOFT	Soft Labels Loss
MAE	Mean Absolute Error
AdamW	Adam with Decoupled Weight Decay
CORN	Conditional Ordinal Regression for Neural networks
CEFR	Common European Framework of Reference for Languages
SAMER	Simplification of Arabic Masterpieces for Extensive Reading
DARES	Dataset for Arabic Readability Estimation of School materials
OSMAN	Open Source Metric for Measuring Arabic Narratives
AARI	Automatic Arabic Readability Index
RANK	Regression plus pairwise margin ranking loss

PLR Regression plus pairwise logistic (RankNet) loss

Abstract

Arabic readability assessment estimates how difficult a text is for a target reader. Readability levels form an ordered scale, yet most systems treat them as unrelated classes and lose the fact that an error of one level is smaller than an error of ten. Before 2025, Arabic readability work relied on formulas adapted from English, on classifiers trained on coarse school or learner scales, and on lexicons and corpora that label words or whole texts; this thesis reviews that work and what it could not support, and builds on the BAREC corpus, which annotates 69,441 Arabic sentences on 19 ordered levels.

The thesis compares ordinal regression heads and loss functions for fine-grained Arabic readability on BAREC. The pipeline stays the same in every experiment: the corpus and its official splits, the morphologically segmented D3Tok input, an AraBERTv2 encoder, and the training protocol of the BAREC benchmark, including its library defaults. Only the prediction head and the loss function change. Nine configurations form the comparable group: cross-entropy, regression with mean squared error, ordinal log loss, an Earth Mover's Distance loss, four soft label settings and a CORAL ordinal head. A two-phase schedule, which warms up with mean squared error and then optimises a differentiable Soft-QWK objective, is reported separately. Three losses that the benchmark does not evaluate are then added from readability research in other languages: CORN, which a BAREC system names as future work, and the pairwise margin ranking and pairwise logistic losses

of multilingual and English, French and Spanish readability models. Three ablations test the learning rate of the CORAL head, the decision threshold and the checkpoint selection rule. A second track freezes the backbone and trains linear heads on cached features over five seeds; it meets the research topic's requirement of being CPU-friendly and gives variance estimates.

The pipeline reproduces the published benchmark: seven of its eight losses land within 1.5 points of quadratic weighted kappa (QWK), and the checkpoint selection rule explains the eighth. Ordinal aware objectives beat the ordinal-blind baseline by up to three QWK points on the development set and four on the test set. Four objectives share the lead: the CORAL head (84.33 test QWK), the two-phase Soft-QWK schedule (84.31), ordinal log loss (84.08) and plain regression (83.64); their differences are a few tenths of a point on a single seed, so the thesis does not name one best loss. When the errors are split by distance, the leaders turn out to gain their QWK by removing far misses and not by lowering the average error. Of the added losses, the margin ranking loss joins the leading group on the test set (84.06), the pairwise logistic loss makes the fewest errors of five levels or more of any run (2.54 per cent on test), and CORN, which removes CORAL's shared projection, trades 1.76 QWK points for 20.24 points of exact accuracy and the lowest mean distance of the study. Two results are about the protocol and not the loss. The cut-points of a CORAL head move less than 0.0703 under the benchmark learning rate, so their initialisation decides where they end up, and this thesis initialises them from the cumulative label frequencies of the training set. The checkpoint selection rule on its own is worth up to +2.00 test QWK points, and is worth most to the methods whose loss is least aligned with the metric.

Keywords: Arabic readability assessment, BAREC, AraBERTv2, D3Tok, ordinal regression, CORAL, CORN, pairwise ranking, Soft-QWK, quadratic weighted kappa, error distance.

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Chapter 1

Introduction

1.1 Background and Motivation

Automatic readability assessment estimates how difficult a text is for a given group of readers. It is an established research topic in Natural Language Processing (NLP) and in educational technology [20, 12], and it is used to choose reading material for learners, to design curricula, to support intelligent tutoring systems and to guide and evaluate text simplification [48]. The field started with readability formulas built on sentence and word length [25, 14], moved to machine learning over hand-crafted linguistic features [43], and is now dominated by pretrained transformer models [36, 17].

Most of that history was written for English. Arabic has received far less attention and is harder to model, because of its rich morphology, its clitics, the usual absence of short vowels in writing, and its lexical ambiguity [39]. Before 2025, Arabic readability work relied on formulas adapted from English [2, 4, 21], on feature-based classifiers trained on school or learner material with three to five levels [42, 30], and on resources such as the SAMER readability lexicon [3], the SAMER simplification corpus [5] and the DARES dataset of Saudi school textbooks [22]. Each of these made progress, but none offered a large, openly split, sentence-level corpus with a fine-grained scale

on which different models could be compared, which Chapter 2 examines in detail.

The Balanced Arabic Readability Evaluation Corpus (BAREC) filled that gap [24]. It contains 69,441 sentences annotated on 19 readability levels, named in the Abjad order of the Arabic letters (1-*alif* to 19-*qaf*) and ranging from kindergarten to postgraduate comprehension. Its release was accompanied by the BAREC 2025 Shared Task [23], in which several systems tried ordinal-aware modelling, among them the ZAI system, which used AraBERTv2 with a CORAL ordinal head [40], and the STBW system, which used a two-phase MSE and Soft-QWK training strategy [46].

Readability levels are ordered. When the true level is 7, predicting level 8 is off by one grade, while predicting level 1 would put a second-grade text at kindergarten level. Standard cross-entropy penalises both errors in the same way and so throws away the ordering of the labels. This is the reason for using ordinal regression methods and distance-aware loss functions [10].

This thesis compares such methods under one identical pipeline: the BAREC corpus and its official splits, the morphologically segmented D3Tok input, an AraBERTv2 encoder and the training protocol of the BAREC benchmark. Only the prediction head and the loss change. Nine configurations form the comparable group: cross-entropy, regression with mean squared error, ordinal log loss, an Earth Mover's Distance loss, four soft-label settings and a CORAL ordinal head. A two-phase Soft-QWK schedule is reported beside them. As our own addition, the thesis then brings to Arabic three loss functions that come from readability research in other languages or that a BAREC system paper names as future work: CORN, the successor of CORAL [44, 40], and two pairwise ranking objectives, the margin ranking loss of a multilingual Wikipedia readability model [47] and the pairwise logistic loss of the neural pairwise ranking model for English, French and Spanish [31]. A second track freezes the encoder and trains only a linear head on cached features. It runs in minutes on a laptop processor, which

satisfies the CPU-friendly setting the research topic asks for, and we repeat it over five seeds so that we can report variance.

1.2 Challenges in Arabic Readability Assessment

Arabic readability assessment is harder than its English counterpart for several linguistic and computational reasons.

The first difficulty is morphology. An Arabic word is often built from a root with prefixes, suffixes and clitics attached, so the same lexeme appears in many surface forms and lexical representations become sparse. Tokenization and feature extraction are therefore harder than in many other languages.

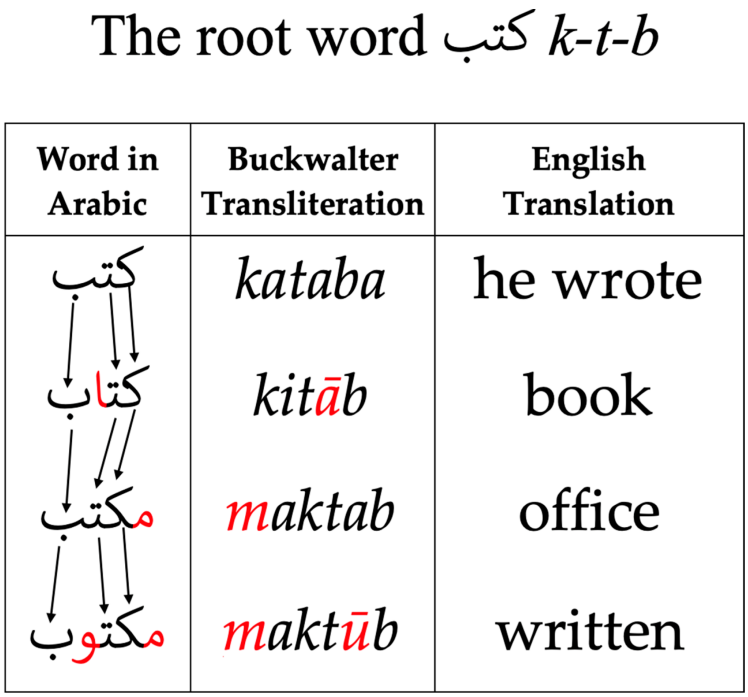


Figure 1.1: Example of Arabic morphological derivation from the root "k-t-b", illustrating the richness and complexity of Arabic word formation.

In addition to root-based derivation, Arabic has extensive *cliticization*: function words such as the definite article, conjunctions, prepositions and pronouns are written attached to a host word. For example, the single written token *walfiḍḍa* ("and the silver") joins three morphological units: the conjunction *wa+* ("and"), the definite article *al+* ("the") and the noun *fiḍḍa* ("silver"). The BAREC annotation guidelines explicitly name the presence and type of clitics as markers of readability level [27], which is why making them explicit through morphological segmentation (D3Tok) is central to the methodology of this thesis.

Lexical ambiguity is another challenge, since a single undiacritized Arabic word can correspond to several readings. The written form *ʿlm*, for example, can be read as *ʿilm* ("knowledge"), *ʿalam* ("flag") or *ʿallama* ("he taught"), depending on the context. Ambiguity affects readability directly, because resolving the rarer readings requires more advanced linguistic knowledge, and the BAREC guidelines place constructions that need diacritization for disambiguation at the higher readability levels. Arabic syntax also allows flexible word order, which makes sentences harder for models to analyse.

Finally, Arabic has few large annotated readability resources compared with English. Earlier Arabic datasets use three to six levels, cover one curriculum, one genre or one learner population, and are annotated at the level of words or whole documents rather than sentences. BAREC changed this situation, but fine-grained prediction across 19 ordered levels remains difficult: two expert annotators agree exactly on only 61.1 per cent of sentences [24].

These challenges lead to two complementary design decisions. First, we use a pretrained transformer encoder whose contextual representations, learned from large Arabic corpora, help with the morphological, lexical and syntactic complexity. Second, we use ordinal-aware learning methods to model the ordered structure of the readability levels, which standard classification ignores.

1.3 Problem Statement

Readability assessment systems often treat readability prediction as a standard classification problem trained with cross-entropy. Readability levels, however, are ordered, and nearby levels are more similar than distant ones. In standard classification, when the true level is 11, predicting level 10 gets exactly the same penalty as predicting level 1, although the first is a near miss and the second a far miss.

This thesis therefore investigates the following questions. Given a fixed AraBERTv2 backbone with D3Tok morphological segmentation and the training protocol of the BAREC benchmark, *which ordinal head and loss function achieves the highest Quadratic Weighted Kappa (QWK) among cross-entropy, regression with mean squared error, ordinal log loss, an Earth Mover's Distance loss, four soft-label settings, a CORAL ordinal head and a two-phase Soft-QWK schedule; how do the loss families differ in the trade-off between exact-match accuracy and distance-aware agreement; and are their errors near misses or far misses?* And, as the contribution that goes beyond the benchmark, *do loss functions developed for readability in other languages, or named as future work by the BAREC systems, carry over to fine-grained Arabic readability?*

1.3.1 Research Objectives

The main objectives of this thesis are:

- Determine which ordinal head and loss function give the highest QWK on the BAREC benchmark when the corpus, the encoder and the entire training protocol stay fixed.
- Compare nine head and loss configurations on a single AraBERTv2 and D3Tok backbone, and report the two-phase Soft-QWK schedule beside them as a separate group.

- Bring to Arabic three loss functions that have not been evaluated on BAREC: CORN and two pairwise ranking objectives from readability research in other languages, each compared with its matched control from the main comparison.
- Quantify the trade-off between exact-match accuracy (*Acc19*) and distance-aware agreement (QWK) across the loss families.
- Measure the error distance of each method, separating near misses from far misses, as the research topic requires.
- Establish whether a CORAL head can learn its cut-points under this protocol at all, and what its cut-points should be initialised to if it cannot.
- Measure how much of the difference between published systems is due to the checkpoint selection rule and the decision threshold rather than to the loss.
- Evaluate predictions at several granularities (19, 7, 5 and 3 levels) using the BAREC collapsing scheme.
- Compare the results with the original BAREC benchmark, with the systems of the BAREC 2025 Shared Task and with the agreement between human annotators.

1.3.2 Research Contributions

The main contributions of this thesis are:

- A review of Arabic readability work before BAREC, from adapted formulas to lexicons, curriculum datasets and learner corpora, that identifies what each one could not support and therefore what a fine-grained sentence-level benchmark adds.

- A controlled comparison of nine head and loss configurations on one fixed pipeline, which extends the four losses of the BAREC benchmark with a CORAL ordinal head, four soft-label settings and a metric-aligned two-phase objective, and reports them in a single table on both official splits.

- The first evaluation on BAREC of three losses taken from outside it: CORN, which the ZAI system paper names as future work [40], and the pairwise margin ranking and pairwise logistic losses used by readability models for Wikipedia in 14 languages [47] and for English, French and Spanish [31].

- A verified reproduction of the benchmark's loss comparison, where the one visible discrepancy is explained and traced back to the checkpoint selection rule.

- An initialisation of the CORAL cut-points from the cumulative training label frequencies, motivated by how far the benchmark learning rate lets them travel, paired with a direct measurement of how much the cut-points actually move and with two ablations that test the argument from opposite directions.

- Two metrics of error distance, the share of predictions at least three and at least five levels away from the gold level. They separate near misses from far misses and show that the lead in QWK comes from removing far misses and not from lowering the average error.

- A measurement of how much the checkpoint selection rule and the CORAL decision threshold are worth, which is what makes systems published under different protocols comparable.

- A track with a frozen backbone that satisfies the CPU-friendly requirement of the research topic, reports variance over five seeds and shows

that the ranking of the losses on frozen features does not carry over to fine-tuning.

1.3.3 Methodology Overview

The system combines a pretrained transformer language model with ordinal-aware learning methods. The pipeline is shown in Figure 1.2.

It begins with the BAREC corpus, where each Arabic sentence carries an ordered readability level from 1 to 19. The input is the D3Tok column of the corpus, a morphologically segmented form in which clitics are separated from their host word and marked with a `+`. That representation is used because AraBERTv2 was pretrained on morphologically segmented Arabic, so segmented input matches its pretraining distribution, and because the pairing of AraBERTv2 with D3Tok is the best of the sixteen model and input combinations the benchmark tested [24]. The segmented text is tokenised and passed to AraBERTv2, and after dropout the pooled output of the encoder, a vector of 768 dimensions, is used as the sentence representation.

This representation goes to a prediction head whose form depends on the configuration. The softmax losses (cross-entropy, ordinal log loss, EMD and the four soft-label settings) use a linear layer with 19 outputs; regression, the Soft-QWK schedule and the two pairwise ranking objectives use a linear layer with a single output; CORAL uses 18 thresholds on one shared projection; and CORN uses 18 independent binary outputs. During training, the corresponding loss compares the output of the head with the gold label. At inference, the output is decoded to a level between 1 and 19 by taking the highest score, by rounding, or by counting how many thresholds are exceeded. Every configuration is trained and evaluated under the same protocol, with QWK as the primary metric, together with exact accuracy (*Acc19*), adjacent accuracy (± 1 *Acc19*), average distance (*Dist*), the collapsed 7-, 5- and 3-level accuracies, and the two error-distance metrics introduced in this thesis.

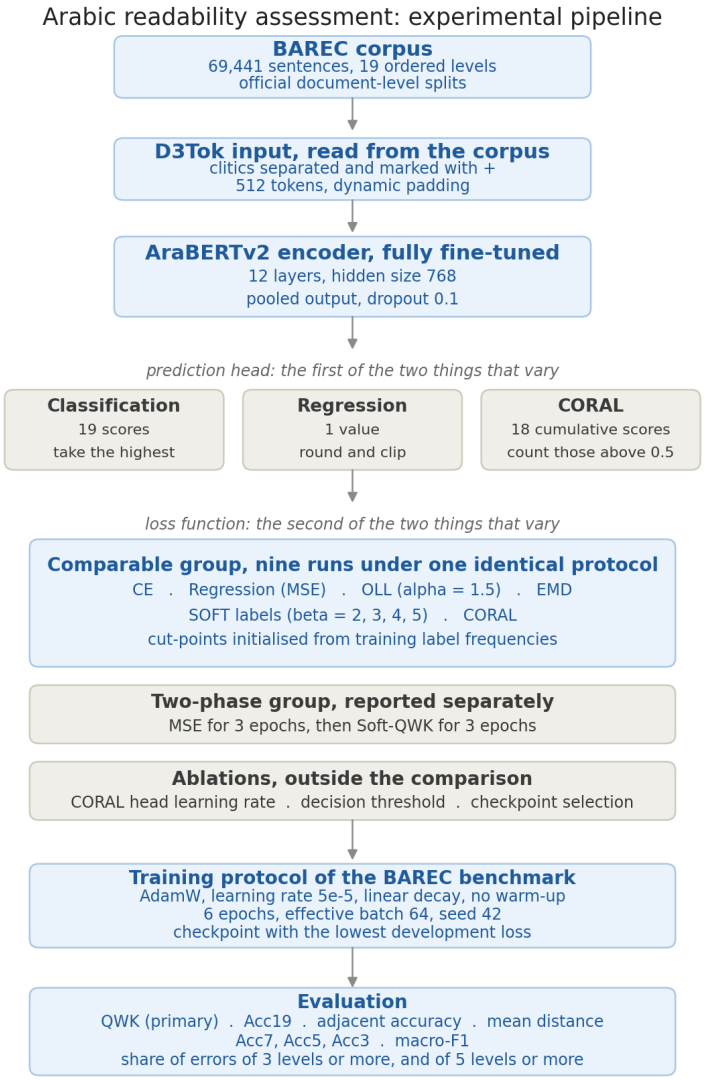


Figure 1.2: The experimental pipeline. Everything drawn in blue is held fixed for every run; the prediction head and the loss function, drawn in grey, are the only components that vary.

1.3.4 Thesis Organization

The thesis is organized as follows.

- **Chapter 1, Introduction:** the background and motivation, the challenges of Arabic readability, the research questions, objectives and contributions, and an overview of the methodology.
- **Chapter 2, Literature Review:** readability assessment in general, from formulas to transformers and to the ordinal and ranking losses used for readability in other languages; then Arabic readability, the Arabic projects that came before BAREC and what each one could not support; then the BAREC corpus, the reasons for the D3Tok input and the AraBERTv2 encoder, the ordinal learning methods, the systems of the BAREC 2025 Shared Task, and where this thesis stands among them.
- **Chapter 3, Proposed Methodology:** the corpus and its splits, the D3Tok input, the encoder and the prediction heads, the exact form of every loss function with the tests that verify it, the three added losses, the initialisation of the CORAL cut-points, the training protocol with every declared difference from the published script, the evaluation metrics and the frozen-backbone track.
- **Chapter 4, Experiments and Results:** the integrity checks, the reproduction of the published benchmark, the full results on both splits, the trade-off between exact accuracy and ordinal agreement, the analysis of error distance, the three ablations of the fine-tuning track and the one of the frozen track, the results of the three added losses, a comparison with published systems and human agreement, and the limitations.
- **Chapter 5, Conclusion and Future Work:** the findings, the contributions, the limitations and the next steps, the first of which is a

study over multiple seeds.

1.4 Summary of Introduction

This chapter introduced readability assessment, the particular difficulties of Arabic and the gap that BAREC filled after two decades of smaller Arabic resources. It stated the research questions, which ask which ordinal head and loss function gives the highest QWK on BAREC when the protocol is held fixed, whether the errors are near misses or far misses, and whether losses from readability research in other languages carry over to Arabic. It listed the nine configurations of the comparable group, the two-phase Soft-QWK schedule, the three added losses and the ablations, and described the AraBERTv2 and D3Tok pipeline and the evaluation metrics.

The next chapter reviews readability assessment in general, then Arabic readability and the projects that preceded BAREC, and then the BAREC corpus and the methods this thesis builds on.

Chapter 2

Literature Review

2.1 Introduction

This chapter moves from the general to the specific. Section 2.2 reviews readability assessment as a research field, from the first formulas to machine learning, deep learning and transformer models, and ends with the ordinal and ranking losses that readability research in other languages has used. Section 2.3 turns to Arabic: what makes it hard and why methods built for English do not transfer directly. Section 2.4 reviews the Arabic readability projects that came before BAREC, formulas, feature-based classifiers, lexicons and corpora, and states for each one what it could not support. Section 2.5 presents BAREC as the answer to those gaps, and Section 2.6 explains the two design choices this thesis inherits from it, the D3Tok input and the AraBERTv2 encoder. Section 2.7 reviews the ordinal learning methods compared in this thesis, Section 2.8 the systems of the BAREC 2025 Shared Task, and Section 2.9 places this thesis among them.

2.2 Readability Assessment

Readability is the ease with which a reader can read and understand a text [20]. Automatic readability assessment predicts it, usually as a grade or a level on an ordered scale, and is used to match texts to readers, to build graded reading material, to evaluate text simplification and to control the difficulty of generated text [12, 48, 52, 1]. Early English datasets were drawn from graded textbooks, and later ones from crowdsourced annotations, simplified news articles and second-language examinations [49, 52, 51, 48].

2.2.1 Traditional Readability Formulas

Early readability research relied on handcrafted formulas that estimate difficulty from shallow counts such as sentence length, word length and syllables per word [20]. The Flesch Reading Ease score [25], the Dale–Chall formula [14], the SMOG grade [37] and the Flesch–Kincaid grade level are the best known. They were developed for English and became popular because they are simple and cheap to compute.

Table 2.1: Examples of traditional readability formulas.

Formula	**Main Features**	**Language**
Flesch Reading Ease	Sentence length and syllables per word	English
Flesch–Kincaid Grade Level	Sentence length and syllables per word	English
Dale–Chall	Sentence length and share of unfamiliar words	English
SMOG Index	Number of polysyllabic words	English

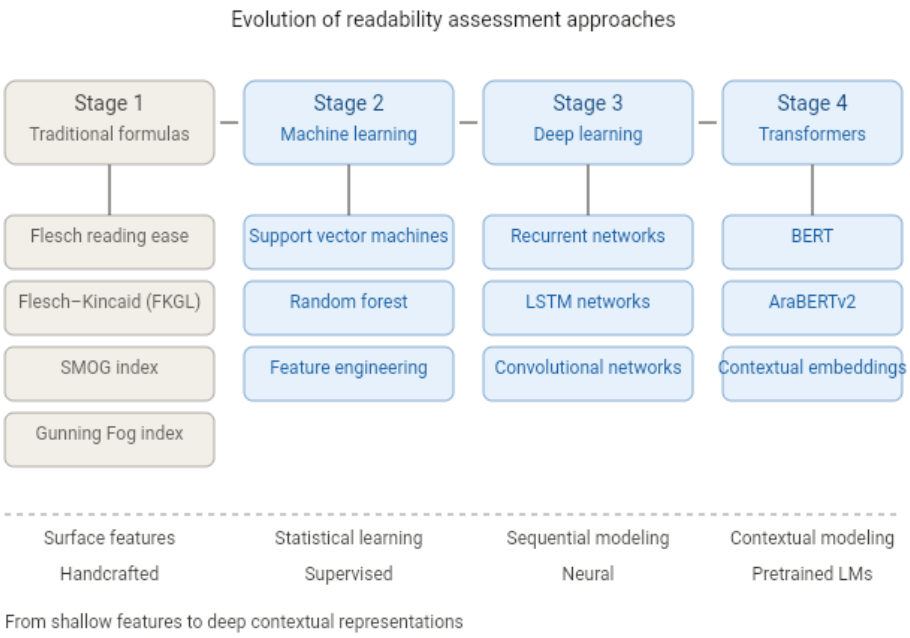


Figure 2.1: How readability assessment evolved, from traditional handcrafted formulas to contextual language models based on transformers.

These formulas assume that longer sentences and longer words make a text harder. They miss vocabulary, syntax, discourse and background knowledge, and they cannot tell apart two texts with the same lengths but very different difficulty. The limitation is sharper for morphologically rich languages such as Arabic, where a single long word may hold a conjunction, an article, a stem and a pronoun [28].

2.2.2 Machine Learning Approaches

Machine learning approaches treat readability as a supervised classification or regression problem over extracted features. Early systems combined lexical, syntactic and statistical features, such as word frequency, the distribution of parts of speech, the depth of the parse tree and language model probabilities,

with classifiers such as support vector machines [43, 13]. Support vector machines, random forests, logistic regression and naive Bayes were all widely used, and they improved clearly on the formulas.

Table 2.2: Common machine learning methods used for readability assessment.

Method	**Main Characteristics**	**Learning Type**
Support Vector Machines	Margin-based supervised classification	Supervised
Random Forest	Ensemble of decision trees	Supervised
Logistic Regression	Linear probabilistic classification	Supervised
Naive Bayes	Probabilistic feature-based classification	Supervised

The BAREC benchmark reports the same family on Arabic: a support vector machine over a count vectorizer on D3Tok input reaches a QWK of only 56.5%, and a decision tree 52.5% [24], far below the pretrained models discussed below.

2.2.3 Deep Learning Approaches

Deep learning removed most of the feature engineering. Recurrent networks, long short-term memory networks, their bidirectional variants and convolutional networks learn representations directly from text, and hierarchical attention networks model the structure of sentences and documents [36]. They capture sequential dependencies and semantic relations that hand-crafted features miss, but they need large annotated datasets and considerable computation when trained from scratch.

Table 2.3: Common deep learning architectures used for readability assessment.

Architecture	**Main Characteristics**	**Learning Type**
RNN	Sequential text modelling	Deep Learning
LSTM	Long-range dependency modelling	Deep Learning
CNN	Local feature extraction	Deep Learning
BiLSTM	Bidirectional contextual sequence modelling	Deep Learning

2.2.4 Transformer-Based Language Models

Transformer models replaced recurrence with self-attention [50], which lets them relate any two words of a sentence directly and process a sequence in parallel. BERT [18] pretrains a bidirectional transformer with masked language modelling on large corpora and is then fine-tuned for the task at hand. Fine-tuned transformers are among the strongest models for readability assessment [36], and Deutsch et al. found that adding hand-crafted linguistic features to them gives little further gain [17].

Table 2.4: Examples of transformer-based language models used in NLP.

Model	**Main Characteristics**	**Language**
BERT	Bidirectional contextual transformer	English
RoBERTa	Optimised BERT pretraining	English
DistilBERT	Compressed transformer	English
AraBERTv2	BERT pretrained on segmented Arabic	Arabic

2.2.5 Ordinal and Ranking Losses in Readability Research

Readability levels are ordered, and research on English and Chinese readability has increasingly used losses that respect that order. Lim and Lee compare ordinal log loss, soft labels, the Earth Mover's Distance and a weighted kappa loss with cross-entropy on English and Chinese corpora, and find that ordinal log loss improves adjacent accuracy significantly on most of them [34]. Zeng et al. train with soft labels for ordinal regression together with a pre-training step that predicts which of two texts is harder [53], and Li et al. weight cross-entropy by the length profile of each level [32].

A second line of work treats readability as ranking. Lee and Vajjala train a neural pairwise ranking model that learns which of two texts is harder, and report zero-shot ranking accuracy above 80 per cent on French and Spanish after training on English [31]; their model is trained with a pairwise logistic loss. Trokhymovych et al. score the readability of Wikipedia articles in 14 languages with a model trained by a margin ranking loss on pairs of articles and their simplified versions [47]. On the ordinal side, CORN extends the CORAL framework by removing its shared weights and training each binary question on a conditional subset of the data [44].

None of these losses was among those evaluated by the BAREC benchmark, and none of the BAREC 2025 system papers that describe their loss reports CORN or a pairwise ranking loss; the ZAI system paper names CORN explicitly as future work [40]. They are therefore the natural candidates for the addition this thesis makes in Chapters 3 and 4.

2.3 Arabic Readability Assessment

Arabic readability is not English readability with different words. Four properties of the language break the assumptions behind English methods [28, 39].

1. **Morphology and clitics.** Words are derived from roots through patterns and carry attached conjunctions, prepositions, articles and pronouns, so word length and word counts mean something different from English, and a word-frequency list sees many forms of one lexeme as different words.

2. **Missing short vowels.** Most Arabic is written without diacritics, so syllables, the basic unit of English formulas, cannot be counted from the written text without first restoring the vowels automatically.

3. **Ambiguity.** An undiacritized word can have several readings, and choosing the rarer one requires more knowledge, which is itself a mark of difficulty.

4. **Diglossia and register.** Modern Standard Arabic coexists with spoken dialects and with Classical Arabic, so the difficulty of a text depends on vocabulary and constructions from very different periods and registers.

These properties mean that an Arabic readability model needs either morphological analysis or a pretrained model that has learned the morphology,

and a dataset whose labels reflect the linguistic phenomena Arabic readers actually find hard. The next section shows how the work before BAREC approached both needs.

2.4 Arabic Readability Projects before BAREC

Arabic readability research before 2025 falls into three groups: formulas adapted from English, feature-based machine learning, and lexical and corpus resources with models built on them [39, 24]. The subsections below describe each group, and Table 2.5 and Section 2.4.5 summarise what they could and could not support. The limitations stated here are limitations for the purpose of this thesis, fine-grained sentence-level modelling; most of these projects were designed for a different purpose and served it well.

2.4.1 Readability Formulas for Arabic

The first Arabic tools followed the English formulas. Al-Khalifa and Al-Ajlan built *Arability*, a prototype that estimates readability from surface features such as word and sentence length with statistical models, using reading books collected manually from the elementary, intermediate and secondary curricula of Saudi schools, that is, three levels [2]. The Automatic Arabic Readability Index (AARI) of Al-Tamimi et al. was fitted by factor analysis on 1,196 texts from the Jordanian curriculum for grades one to ten, and combines the number of characters, the average word length and the average sentence length in a single linear formula [4].

OSMAN, by El-Haj and Rayson, adapts the Flesch approach to Arabic [21]. Because Arabic is usually written without short vowels, it first adds diacritics automatically with an open-source diacritizer and then counts short, long and stressed syllables; it also adds a *Faseeh* factor for letters and word

endings typical of formal writing. It was evaluated by correlating its scores over 73,000 parallel English and Arabic sentences from United Nations documents.

These formulas share the weaknesses of their English models and add some of their own. They use surface counts that ignore morphology, syntax and vocabulary, which are exactly the dimensions along which Arabic difficulty varies. They were calibrated on the school curriculum of one country, so their scale is the grade system of that curriculum. OSMAN's syllable counts depend on the accuracy of an automatic diacritizer, and its evaluation goes through the parallel English sentences rather than through levels assigned to Arabic text by Arabic readers. None of them produces a fine-grained level that can be checked against expert labels at sentence level.

2.4.2 Feature-Based Machine Learning

Saddiki et al. studied 146 features, from average word length to the depth of the syntactic parse tree, for Arabic as a first and as a second language [42]. On a first-language corpus of 27,688 texts in four levels their best model reaches 94.8 per cent accuracy, and on the second-language task 72.4 per cent. Khallaf and Sharoff classified the difficulty of Modern Standard Arabic sentences for language learners, either on a three-way scale derived from the Common European Framework of Reference (CEFR) or as simple against complex [30]. They compared sentence embeddings from fastText, multilingual BERT, XLM-R and an Arabic BERT with traditional features, and obtained their best result, an F1 of 0.80 on the three-way task and 0.94 on the binary task, with the fine-tuned Arabic BERT.

These studies showed that learned models beat formulas and that pre-trained Arabic encoders help, but their high scores are obtained on coarse scales of two to four levels, where a large share of the task is separating very easy from very hard text. The first works on whole texts, not sentences, and the second on a learner scale collapsed to three classes. Neither scale can

express the difference between, say, a text for the second and one for the third school grade.

2.4.3 Lexical Resources and Corpora

The SAMER project took a lexical route. Its readability lexicon assigns one of five levels to about 26,000 lemmas of Modern Standard Arabic, each annotated in triplicate by language professionals from three regions of the Arab world [3]. It led to the SAMER simplification corpus, the first manually annotated Arabic parallel corpus for text simplification: 159,000 words from 15 Arabic novels, most of them published between 1865 and 1955, each simplified by hand to two lower levels [5]. In that corpus the level of a text is defined as the highest level of any of its words. Liberato et al. used it to model readability at the level of words and of text fragments, combining frequency, rule-based and pretrained models, and reached a macro F1 of 86.7 at word level and 87.9 at fragment level on a blind test set [33].

DARES collects 13,335 texts from the Saudi school textbooks of 2021, with a coarse task that predicts the educational stage and a fine task that predicts the school grade [22]. ReadMe++ annotates 9,757 sentences in Arabic, English, French, Hindi and Russian from 112 sources and 21 domains on the six CEFR levels [38], so Arabic is one of five languages in it.

Each resource answered a real need, and each carries a limitation for fine-grained modelling. The SAMER lexicon rates words out of context, and the SAMER corpus inherits that view: a sentence is as hard as its hardest word, whatever its syntax, and the texts are nineteenth- and early-twentieth-century fiction. DARES labels a text with the grade of the textbook it came from, which is a property of where the text was published rather than a judgement of the text itself, and it covers one country's curriculum in one year. ReadMe++ uses a second-language scale of six levels, built for comparison across languages rather than for the Arabic school system.

2.4.4 Summary of the Projects

Table 2.5: Arabic readability work before BAREC. The last column states the main limitation for fine-grained, sentence-level modelling, which is not always the purpose the work was designed for.

Work	**Year**	**Unit**	**Scale**	**Data**	**Main limitation**
Arability [2]	2010	text	3 levels	Saudi school reading books	surface features, one curriculum
AARI [4]	2014	text	grades 1–10	1,196 Jordanian curriculum texts	linear formula of length counts
OSMAN [21]	2016	text	continuous score	73,000 UN sentence pairs	depends on automatic diacritization; validated through English
Saddiki et al. [42]	2018	text	4 levels	27,688 texts	coarse scale, heavy feature pipeline
SAMER lexicon [3]	2020	word	5 levels	about 26,000 lemmas	words out of context
Khallaf and Sharoff [30]	2021	sentence	3 CEFR levels	learner sentences	learner scale, coarse
SAMER corpus [5]	2024	word, text	5 levels	159,000 words, 15 novels	one genre; level = hardest word
Liberato et al. [33]	2024	word, fragment	SAMER levels	SAMER corpus	inherits SAMER's lexical definition
DARES [22]	2024	text	stages, grades	13,335 textbook texts	label is the textbook's grade
ReadMe++ [38]	2024	sentence	6 CEFR levels	9,757 sentences, 5 languages	Arabic is one fifth; L2 scale

2.4.5 What Was Missing

Read together, the projects leave six gaps for anyone who wants to model fine-grained Arabic readability and to compare models on it.

1. **Granularity.** The scales range from three levels to one level per school grade. A model evaluated on three levels can be very accurate while being unable to separate adjacent school grades.

2. **Unit.** The labels are attached to words or to whole texts. Sentence-level labels, which let a model be checked against a human judgement for every sentence, exist only in learner-oriented or multilingual form.

3. **Source of the label.** Several labels are proxies: the grade of the source textbook, the hardest word of a text, or the readability of a translation. None is an expert's judgement of the specific text against explicit linguistic criteria.

4. **Coverage.** Each resource covers one curriculum, one genre or one learner population.

5. **Features.** The formulas and most feature sets use length counts and ignore the morphology that makes Arabic hard.

6. **A shared benchmark.** Without one large corpus with fixed public splits and an agreed metric, results obtained on different datasets cannot be compared.

2.5 The BAREC Corpus

The Balanced Arabic Readability Evaluation Corpus was built to close these gaps [24, 27]. Its levels follow the 19-level system of Taha-Thomure for Arabic children's books [45], named in the Abjad order of the Arabic letters from

1-*alif* to 19-*qaf* and running from kindergarten to postgraduate comprehension, with finer distinctions at the lower levels where readability varies most. Through the BAREC pyramid the 19 levels collapse into coarser schemes of 7, 5 and 3 levels, so one model trained at the finest granularity can be evaluated at any coarser one.

Annotators assign each sentence the level of its most difficult linguistic phenomenon, judged along six dimensions: spelling, word count, morphology, syntax, vocabulary and content. The guidelines define which phenomena of each dimension qualify a sentence for a given level; clitics and infrequent inflections such as the dual, for example, are morphological markers of higher levels [27]. The annotation team consisted of six native speakers, all experienced Arabic language educators, and every sentence of the first phase was checked twice.

The corpus contains 69,441 sentences and about one million words from 1,922 documents drawn from 30 resources, organised in three domains (Arts and Humanities, Social Sciences, STEM) and three readership groups (Foundational, Advanced, Specialized); about a quarter of the sentences come from sources that were typed by hand to make them digitally usable [24]. It is split at the document level into training, development and test sets of 80, 10 and 10 per cent, so that no document contributes to more than one split.

Agreement between annotators was measured in 16 studies. The average pairwise exact agreement on the 19 levels is only 61.1 per cent, but the average pairwise QWK is 81.8 per cent and the average distance 0.94 levels, so most disagreements are between neighbouring levels. Against the unified labels agreed after discussion, individual annotators reach 71.7 per cent exact agreement and a QWK of 88.1 per cent [24].

Table 2.6: Main characteristics of the BAREC corpus [24].

Characteristic	**Description**
Language	Modern Standard Arabic
Size	69,441 sentences, about 1 million words
Sources	1,922 documents from 30 resources
Domains	Arts and Humanities, Social Sciences, STEM
Readership groups	Foundational, Advanced, Specialized
Readability levels	19 ordered levels, collapsible to 7, 5 and 3
Unit of annotation	sentence
Annotation	manual, by six expert Arabic educators, over six linguistic dimensions
Agreement	81.8% pairwise QWK; 88.1% QWK against the unified label
Splits	80% train, 10% development, 10% test, at document level

Table 2.7 shows how the corpus answers each gap of Section 2.4.5.

Table 2.7: The gaps left by earlier Arabic readability work and the corresponding property of BAREC.

Gap	**Before BAREC**	**BAREC**
Granularity	from 3 levels to one per school grade	19 levels, collapsible to 7, 5 and 3
Unit	words or whole texts	every sentence labelled
Source of the label	textbook grade, hardest word, English side	expert judgement against explicit criteria in six dimensions
Coverage	one curriculum, genre or learner group	30 resources, 3 domains, 3 readership groups
Measured reliability	rarely reported at sentence level	16 agreement studies, 81.8% pairwise QWK
Shared benchmark	different datasets and metrics	fixed document-level splits, QWK, a shared task

The benchmark paper that introduced the corpus also gives the reference results that this thesis reproduces. The best single model, AraBERTv2 with D3Tok input trained as a regression, reaches 84.4% QWK on the test set; an average of six models reaches 84.5%; and an oracle that picks the best of the six predictions for each sentence reaches 94.7% [24].

2.6 Input Representation and Encoder

Two components of the pipeline are chosen once and kept fixed: the form in which a sentence is given to the model, and the pretrained encoder that reads it. Both follow the evidence in the BAREC benchmark.

2.6.1 D3Tok: Morphological Segmentation

The corpus provides four input variants of every sentence, produced with the morphological disambiguator of CAMeL Tools [41, 24]. *Word* is the tokenised, dediacritized sentence; *Lex* replaces every word by its lemma;

D3Tok splits every word into its base and its clitics and marks each clitic boundary with a `+`; and *D3Lex* replaces the base of the D3Tok form by its lemma. The written token *walfiḍḍa* ("and the silver"), for instance, becomes `w+ Al+ fDp` in D3Tok, with the conjunction, the article and the noun visible as separate units.

D3Tok is the right choice for readability for two reasons. Linguistically, clitics are among the features the annotation guidelines use to set a sentence's level [27], so the segmentation shows the encoder the very units the annotators looked at, while keeping the word forms that Lex and D3Lex discard. Empirically, D3Tok with AraBERTv2 is the best of the sixteen model and input combinations the benchmark evaluated, as Table 2.8 shows.

2.6.2 AraBERTv2

AraBERTv2 [6] is a BERT-base model (12 layers, 12 attention heads, hidden size 768) pretrained on about 8.6 billion words of Arabic. What sets it apart from the other Arabic encoders is its pretraining data: the text was segmented with the Farasa segmenter [15], which also separates clitics from their host word and marks the boundaries with `+`. D3Tok input therefore matches the distribution AraBERTv2 saw during pretraining, which is not true of any other encoder in the comparison.

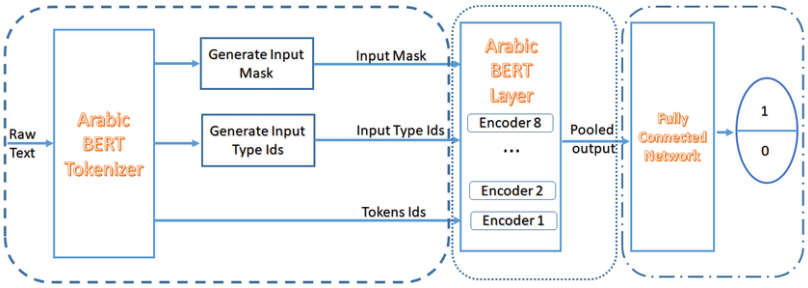


Figure 2.2: General architecture of transformer-based Arabic language models such as AraBERTv2.

Table 2.8 reproduces the evidence from the benchmark paper. AraBERTv2 is the only encoder that improves when the input is segmented: its QWK rises from 76.3% with word input to 80.0% with D3Tok, while the three other encoders lose between 0.2 and 1.0 points. The benchmark therefore carries AraBERTv2 with D3Tok into its comparison of loss functions, and this thesis does the same so that its reproduction of that comparison is meaningful.

Table 2.8: Development-set QWK (%) of four Arabic encoders on the four input variants, all trained with cross-entropy, as reported in Table 5 of the BAREC benchmark [24]. The best value is in bold.

Encoder	**Word**	**Lex**	**D3Tok**	**D3Lex**
CamelBERT-msa	79.1	77.1	78.2	78.0
MARBERTv2	79.1	77.0	78.9	78.0
AraBERTv02	79.2	78.5	78.2	78.2
AraBERTv2	76.3	77.7	**80.0**	78.6

2.7 Ordinal Regression and Distance-Aware Learning

Ordinal regression is the family of methods for labels with a natural order [26]. Where classification treats the classes as unrelated, ordinal methods take the distance between them into account, which suits readability: predicting level 10 instead of 11 should cost less than predicting level 1 instead of 11.

Table 2.9: Ordinal-aware learning methods compared in this thesis.

Method	**Main Characteristics**	**Paradigm**
Cross-Entropy (CE)	Classification without ordinal awareness	Classification
Regression (MSE)	Continuous prediction with a squared-error loss	Regression
CORAL	$K-1$ binary questions with one shared projection	Ordinal regression
CORN	$K-1$ conditional binary questions, no shared weights	Ordinal regression
Ordinal Log Loss (OLL)	Distance-weighted penalty on wrong levels	Ordinal classification
Earth Mover's Distance (EMD)	Squared difference of cumulative distributions	Ordinal classification
Soft labels (SOFT)	Targets that decay with distance	Ordinal classification
Soft-QWK	Differentiable surrogate of the QWK metric	Metric optimisation
Pairwise ranking (RANK, PLR)	Regression plus a penalty on misordered pairs	Learning to rank

CORAL models the ordinal prediction as a sequence of $K-1$ binary questions "is the level greater than k?" that share one projection and differ only in their bias, so the ordering is built into the output [9]. CORN keeps the $K-1$ questions but gives each its own weights and trains question k only on the examples above level $k-1$, recovering the probabilities by the chain rule [44]. Ordinal log loss weights the probability given to each wrong level by its distance from the true level [10]; the soft-label loss replaces the one-hot target by one that decays with distance [7]; and the Earth Mover's Distance loss compares cumulative distributions [29]. The Soft-QWK loss optimises a differentiable surrogate of QWK [16, 19]; because that surrogate is a statistic of the whole batch, it can be lowered by predicting near the mean level, and the STBW system therefore precedes it with a mean squared error warm-up

[46]. The pairwise ranking losses of Section 2.2.5 add a penalty whenever a harder text is scored below an easier one, either with a hinge [47] or with a logistic function [31, 8].

2.8 Systems Developed for the BAREC Shared Task

The BAREC 2025 Shared Task on Arabic readability assessment was organised at the Third Arabic Natural Language Processing Conference (ArabicNLP 2025) [23]. It attracted 22 registered teams from 12 countries, and 17 of them submitted system description papers. It defined two subtasks, at sentence level and at document level, and three resource tracks, Strict, Constrained and Open, with QWK on a blind test set as the official ranking metric.

The most successful systems combined D3Tok segmentation with Arabic pretrained encoders, ordinal-aware losses and ensembling. Large language models and less standard architectures such as graph neural networks and vision–language models generally did worse than the transformer baselines, and only five teams went above the strongest baseline of the organisers (AraBERTv2 with D3Tok and regression, 84.6% blind QWK).

Two systems are especially relevant to this thesis. The ZAI system used AraBERTv2 with a CORAL head trained with a binary cross-entropy summed over the $K-1$ thresholds; it selected its checkpoint by QWK, stopped early with a patience of two epochs and tuned a single decision threshold on the development set, reaching 85.5% QWK on the blind test set [40]. Its paper names alternative ordinal objectives such as CORN as future work. The STBW system used AraBERTv2 with a single-output regression head and a two-phase schedule, a mean squared error warm-up followed by a differentiable Soft-QWK objective, which raised its own test QWK from 83.9% to 84.88% and reached 85.6% on the blind test set; besides the seg-

mented sentence, its input carries corpus metadata and surface statistics [46]. These two systems are the reason both CORAL and Soft-QWK are included in the present comparison, and because their protocols differ, this thesis reimplements both inside one fixed pipeline instead of comparing published numbers.

Table 2.10 summarises the benchmark, the baselines of the organisers and the participating systems, with their approaches and their sentence-level QWK.

Table 2.10: Work on the BAREC dataset: corpus benchmark, organiser baselines, and BAREC 2025 Shared Task systems (sentence level, Strict track). QWK is the primary metric; higher is better.

Team / Source	**Year**	**Model / Approach**	**Loss / Method**	**QWK (%)**
Corpus benchmark (Elmadani et al., 2025), test set				
BAREC best single	2025	AraBERTv2 + D3Tok	Regression (MSE)	84.4
BAREC ensemble	2025	Average of 6 AraBERTv2 models	Ensemble	84.5
BAREC oracle	2025	Best of 6 per sentence	Oracle	94.7
SVM baseline	2025	SVM, count vectorizer, D3Tok	Classification	56.5
Organiser baselines, blind test set				
Baseline III	2025	AraBERTv2 + D3Tok	Regression	84.6
Baseline II	2025	AraBERTv2 + D3Tok	CE	81.5
Baseline I	2025	AraBERTv02 + Word	CE	80.5
Shared Task systems, blind test set (Strict track, sentence level)				
!MSA	2025	Ensemble of 4 Arabic PLMs + LLM augmentation	CE + OLL + MSE + ordinal	**87.5**
AMAR	2025	Stacked meta-learner over 7 transformers	Regression + meta-classifier	86.4
mucAI	2025	AraBERTv2 + conformal prediction	CE / EMD / Focal	85.7
STBW	2025	AraBERTv2, two-phase training	MSE → Soft-QWK	85.6
ZAI	2025	AraBERTv2 + CORAL ordinal head	CORAL (BCE)	85.5*
Syntaxa	2025	AraBERTv2 + dependency-graph GNN	GNN	84.3
MorphoArabia	2025	Morphology-aware regression	Regression	84.2
MARSAD	2025	Specialised AraBERT, two-stage	CE → WKL	84.1
Noor	2025	Hybrid transformer + features	PLM + features	83.1
Phantoms	2025	Hybrid BERT + linguistic features	PLM + features	82.7
Qais	2025	Fine-tuned PLM (+ LLM)	Classification	82.5
LIS	2025	Curriculum learning with	Curriculum	82.4

2.9 Positioning of This Thesis

This thesis has a different aim from the shared-task systems. It does not try to maximise the leaderboard through ensembling, data augmentation or external resources. It isolates the effect of the prediction head and the loss function on one fixed AraBERTv2 and D3Tok pipeline, with every other component kept at the value the benchmark script sets, and it adds to the benchmark's losses three kinds of objective that the benchmark does not evaluate: the ordinal heads used by the strongest systems (CORAL and the two-phase Soft-QWK schedule), the successor of CORAL that one of those systems names as future work (CORN), and the pairwise ranking losses of readability research in other languages. Every one of them is compared with a matched control under the same protocol, so that a difference in the results can be attributed to the loss.

2.10 Summary

This chapter traced readability assessment from formulas based on sentence and word length to transformer models, and showed that research on English and other languages has moved towards losses that respect the order of the levels or treat readability as ranking. It then described what makes Arabic different and reviewed the Arabic work before BAREC: formulas such as Arability, AARI and OSMAN; feature-based classifiers such as those of Saddiki et al. and Khallaf and Sharoff; and resources such as the SAMER lexicon and corpus, DARES and ReadMe++. Each of them worked on coarse scales, on words or whole texts, on one curriculum, genre or learner population, or with labels that are proxies for difficulty. BAREC answers these gaps with 69,441 expert-labelled sentences on 19 levels, fixed splits and measured agreement, and its benchmark shows why AraBERTv2 with D3Tok input is the right fixed backbone. The chapter ended with the ordinal learning methods and

the shared-task systems that this thesis builds on, and with the position of this thesis among them.

The next chapter presents the methodology: the corpus and its splits, the D3Tok input, the encoder and the prediction heads, the exact form of every loss function including the three added ones, the initialisation of the CORAL cut-points, the training protocol with its declared differences from the published script, and the evaluation setup.

Chapter 3

Proposed Methodology

3.1 Introduction

This chapter describes how the comparison announced in Chapter 1 is carried out. The question is narrow on purpose: on one fixed corpus, one fixed encoder and one fixed training protocol, what changes when only the prediction head and the loss function change? Answering it requires that everything else stay identical, so most of this chapter is concerned with what was held fixed and with how that was verified.

Section 3.2 states the design and the two experimental tracks. Section 3.3 describes the BAREC corpus, the official splits and the checks run on the loaded data. Section 3.4 describes the D3Tok input representation. Section 3.5 describes the encoder and the three prediction heads. Section 3.6 gives the exact form of every loss function, together with the numerical safeguards and the tests that verify each implementation. Section 3.7 describes the three losses added from readability research in other languages. Section 3.8 presents the cut-point initialisation for CORAL, which is a contribution of this thesis. Section 3.9 states the training protocol and every point at which the implementation departs from the published BAREC script. Section 3.10 lists the methods compared and the ablations. Section 3.11 defines

the evaluation metrics. Section 3.12 describes the frozen-backbone track that answers the CPU-friendly requirement of the research brief, and Section 3.13 reports the implementation, the hardware and the reproducibility record.

3.2 Experimental Design

The experiments follow the pipeline of the BAREC benchmark [24], whose training script is treated here as the specification of the protocol. The corpus, the input representation, the encoder, the optimiser, the learning rate, the number of epochs, the batch size, the schedule, the random seed and the checkpoint selection rule are the same for every method. Two components vary: the prediction head, which turns a sentence representation into either 19 scores, one continuous value or 18 cumulative scores, and the loss function applied to that output. This isolation is the reason the comparison can be attributed to the ordinal mechanism rather than to a difference in tuning.

Two tracks are reported.

1. **Fine-tuning track (main results).** The whole encoder is fine-tuned, exactly as in the BAREC benchmark. This track is the only one whose numbers are comparable with the published tables, and it carries the answer to the research question.

2. **Frozen-backbone track.** The encoder is run once in inference mode, its sentence features are cached, and only a linear head is trained on those cached features. Head training takes minutes on a laptop processor, which satisfies the CPU-friendly setting requested in the research brief, and it is cheap enough to repeat over five random seeds, so every number in that track is reported with a standard deviation. Because the encoder never adapts to the task, these numbers are far below the fine-tuning track and are never compared with it or with the published tables.

A deliberate limitation of the fine-tuning track should be stated at the outset: each method is trained once, with the seed fixed by the reference protocol at 42. The frozen track is the part of this work that reports variance.

3.3 Data

3.3.1 The BAREC Corpus

All experiments use the Balanced Arabic Readability Evaluation Corpus [24], obtained from the public repository `CAMeL-Lab/BAREC-Corpus-v1.0`. The corpus contains 69,441 manually annotated Arabic sentences over 19 ordered readability levels, named after the Abjad ordering of the Arabic letters (1-*alif* to 19-*qaf*) and ranging from kindergarten to postgraduate reading difficulty. Sentences carry the fine-grained label in the column `Readability_Level_19` and the three collapsed labels `Readability_Level_7`, `Readability_Level_5` and `Readability_Level_3`.

The corpus is split at the document level, so that no document contributes sentences to more than one split. Table 3.1 reports the sizes.

Table 3.1: BAREC splits as used in this thesis. The sizes are asserted against Table 3 of Elmadani et al. [24] before any training starts.

Split	**Documents**	**Sentences**	**Words**
Train	1,518	54,845	832,743
Development	194	7,310	101,364
Test	210	7,286	105,264
All	1,922	69,441	1,039,371

Labels are shifted to $y \in \{0, \ldots, 18\}$ during training and shifted back only when results are reported, which follows the reference script and keeps the index arithmetic in the losses simple.

3.3.2 Checks Run on the Loaded Data

Four assertions run before any model is built, and the program stops rather than continue if one of them fails.

1. The three split sizes equal the published values in Table 3.1.
2. The D3Tok column exists, and the clitic marker `+` occurs in more than half of a 2,000-sentence sample of the training split. A silent preprocessing failure would leave plain text in that column, and this check catches it.
3. The 7-, 5- and 3-level collapse tables of the reference script are applied to the 19-level gold labels and compared, sentence by sentence, with the corpus's own collapsed columns on all three splits. They agree everywhere, so the collapsed accuracies reported in Chapter 4 cannot be the product of a misremembered mapping.
4. The development and test splits each contain at least one sentence of every one of the 19 levels, which the quadratic weighted kappa computation requires to be meaningful.

One correction to the published script is needed: the released dataset names its middle split `dev`, while the script reads a split named `validation`. The implementation accepts either name and then verifies the size, so the correct split is used.

3.4 Input Representation

3.4.1 D3Tok

The corpus ships several input variants of every sentence. This thesis uses D3Tok, the morphologically segmented variant produced by the corpus authors with CAMeL Tools [41]: each word is separated into its base and its

clitics, and every clitic boundary is marked with a `+` character. D3Tok is read as a column of the corpus, so the segmentation is the corpus authors' own and is identical for every method, which removes the preprocessing tool as a source of variation between methods.

Arabic attaches function words to their host word in writing. A single written token may fuse a conjunction, the definite article and a noun; in Buckwalter transliteration such a token appears as `wAlfDp` and D3Tok renders it `w+ Al+ fDp`, exposing the conjunction *wa+*, the article *al+* and the noun. The BAREC annotation guidelines list the presence and the type of clitics among the markers of readability level [27], so exposing them to the encoder is linguistically motivated and not only a tokenisation convenience.

Two empirical reasons support the choice. First, AraBERTv2 was pretrained on morphologically segmented Arabic [6], so segmented input matches the distribution seen during pretraining. Second, in the benchmark itself, the pairing of AraBERTv2 with D3Tok input raises development QWK from 76.3 to 80.0 relative to plain word input, and that pairing is the configuration the benchmark carries into its loss function comparison [24]. Using it here is what makes the reproduction check in Chapter 4 meaningful.

3.4.2 Tokenisation

Sentences are tokenised with the AraBERTv2 tokenizer, truncated at 512 subword tokens, and padded dynamically to the longest sequence in each batch. Truncation length and dynamic padding both follow the reference script. Dynamic padding changes only how much padding the encoder processes, never the content of a sentence.

3.5 Encoder and Prediction Heads

3.5.1 Encoder

The encoder is AraBERTv2 (`aubmindlab/bert-base-arabertv2`) [6]: twelve transformer layers, twelve attention heads and a hidden size of 768, pretrained on approximately 8.6 billion words of Arabic. The same checkpoint, at the same repository revision, is used by every method in the fine-tuning track, and all of its parameters are updated during training.

The sentence representation is the encoder's pooled output, that is, the first-token representation passed through the pretrained pooling layer with its hyperbolic tangent activation. A dropout layer with the rate stored in the model configuration, 0.1, is applied to it. This is the representation that the standard sequence classification wrapper uses, so the classification and regression heads take it unchanged; the CORAL head is built on the same representation so that the three heads differ in their output structure only.

3.5.2 Three Heads

Let $\mathbf{h} \in \mathbb{R}^{768}$ be the pooled, dropped-out representation and let $K = 19$.

Classification head. A linear layer produces one score per level,

$$\mathbf{z} = W\mathbf{h} + \mathbf{b}, \qquad W \in \mathbb{R}^{19\times 768}, \quad \mathbf{b} \in \mathbb{R}^{19}, \tag{3.1}$$

which a softmax turns into a distribution $\mathbf{p}$ over the 19 levels. Prediction is $\hat{y} = \arg\max_k p_k$. Cross-entropy, OLL, EMD and the four soft-label losses all use this head; they differ only in the loss applied to $\mathbf{p}$.

Regression head. A linear layer produces one unbounded scalar,

$$\hat{s} = \mathbf{w}^\top \mathbf{h} + b, \qquad \mathbf{w} \in \mathbb{R}^{768}, \quad b \in \mathbb{R}. \tag{3.2}$$

Prediction rounds to the nearest level and clips to the valid range, $\hat{y} = \text{clip}(\text{rint}(\hat{s}), 0, 18)$, where rint rounds halves to the nearest even integer. The mean squared error method and the Soft-QWK method use this head.

CORAL head. The CORAL head [9] replaces the 19-way decision by $K - 1 = 18$ ordered binary questions, "is the level greater than k?". All 18 questions share a single projection vector and differ only in a free bias, called a cut-point:

$$z_k = \mathbf{w}^\top \mathbf{h} + b_k, \qquad k = 0, \ldots, 17, \tag{3.3}$$

implemented as a bias-free linear layer of output size one, giving a single readability score $s = \mathbf{w}^\top \mathbf{h}$, plus a learnable vector of 18 cut-points added to it. After a sigmoid, $\sigma(z_k)$ estimates $P(y > k)$. Prediction counts the questions answered yes,

$$\hat{y} = \sum_{k=0}^{17} \mathbb{1}\left[\sigma(z_k) > 0.5\right]. \tag{3.4}$$

Because the projection is shared, the 18 estimated probabilities are ordered whenever the cut-points are ordered, so the model cannot claim that a sentence exceeds level 9 while denying that it exceeds level 5. Cao et al. prove that the minimiser of the CORAL loss has ordered cut-points [9]; that is a statement about the optimum, not a guarantee about every intermediate model, so whether the cut-points are in fact ordered after each epoch is logged during training and reported in Chapter 4.

The three heads add 14,611, 769 and 786 parameters respectively on top of the encoder, so the capacity added by the head is negligible against the roughly 135 million parameters of AraBERTv2 and cannot explain any difference between methods.

Table 3.2 summarises the heads and their decoding rules.

Table 3.2: Prediction heads, their outputs and their decoding rules.

Head	**Output**	**Used by**	**Decoding**
Classification	$\mathbb{R}^{19}$	CE, OLL, EMD, SOFT ($\beta = 2, 3, 4, 5$)	$\arg\max_k p_k$
Regression	$\mathbb{R}$	Regression (MSE), MSE $\rightarrow$ Soft-QWK	round to nearest level, clip to $[0, 18]$
CORAL	$\mathbb{R}^{18}$	CORAL and its ablation	count of $\sigma(z_k) > 0.5$

3.6 Loss Functions

Throughout this section $y \in \{0, \ldots, 18\}$ is the gold level of one sentence, $\mathbf{p}$ is the predicted distribution from the classification head, $\hat{s}$ is the output of the regression head, and N is the number of sentences in the batch. Every loss is a mean over the batch, which is the reduction used by the reference script.

3.6.1 Cross-Entropy

$$\mathcal{L}_{\text{CE}} = -\log p_y. \tag{3.5}$$

Cross-entropy is the ordinal-blind baseline. Its value depends only on the probability given to the correct level, so moving the remaining mass from an adjacent level to the most distant level does not change it. Every other loss in this comparison exists to remove that property.

3.6.2 Regression with Mean Squared Error

$$\mathcal{L}_{\text{MSE}} = (\hat{s} - y)^2. \tag{3.6}$$

The penalty grows with the square of the error, so a prediction ten levels away costs one hundred times a prediction one level away. Ordinal information enters through the arithmetic of the label rather than through an

explicit distance matrix. This is the method that obtained the best QWK in the benchmark [24].

3.6.3 Ordinal Log Loss

Ordinal log loss [10] keeps the classification head and weights the probability given to each wrong level by its distance from the gold level:

$$\mathcal{L}_{\text{OLL}} = -\sum_{k=0}^{18} |y-k|^{\alpha} \log\left(1-p_k\right), \qquad \alpha = 1.5. \tag{3.7}$$

The gold level contributes nothing, since its weight $|y-y|^{\alpha}$ is zero; the loss is driven entirely by where the incorrect mass sits. The exponent α controls how sharply distant mass is punished relative to adjacent mass, and $\alpha = 1.5$ is the value that performed best among $\{0.5, 1, 1.5, 2\}$ in the benchmark, which is why it is the value used here.

3.6.4 Earth Mover's Distance

The EMD loss [29] compares cumulative distributions rather than probabilities. With $P_k = \sum_{j \le k} p_j$ and the gold cumulative distribution $T_k = \mathbb{1}[y \le k]$,

$$\mathcal{L}_{\text{EMD}} = \sum_{k=0}^{18} \left(P_k - T_k\right)^2. \tag{3.8}$$

The sum is not divided by the number of levels, again following the reference implementation. Probability mass that sits near the gold level produces a short mismatch between the two cumulative curves, and mass that sits far from it produces a long one.

3.6.5 Soft Labels

The soft-label loss [7] replaces the one-hot target by a distribution that decays exponentially with distance from the gold level, and then applies cross-entropy against that target:

$$q_i = \frac{e^{-\beta|y-i|}}{\sum_{n=0}^{18} e^{-\beta|n-i|}}, \qquad \mathcal{L}_{\text{SOFT}} = -\sum_{i=0}^{18} q_i \log p_i. \tag{3.9}$$

The normalisation runs over the levels n for a fixed i, as in the reference implementation, so the target is not exactly a probability vector over i; it is a distance-weighted target whose scale depends on β. The smoothing parameter is evaluated at $\beta \in \{2, 3, 4, 5\}$, the grid used in the benchmark. A large β concentrates the target on the gold level and approaches cross-entropy; a small β spreads credit onto neighbouring levels.

3.6.6 CORAL

For the CORAL head, the target of question k is $t_k = \mathbb{1}[y > k]$ and the loss is the binary cross-entropy of the 18 questions, summed over questions and averaged over the batch:

$$\mathcal{L}_{\text{CORAL}} = \frac{1}{N} \sum_{n=1}^{N} \sum_{k=0}^{17} \Big[-t_k^{(n)} \log \sigma\big(z_k^{(n)}\big) - \big(1 - t_k^{(n)}\big) \log \big(1 - \sigma\big(z_k^{(n)}\big)\big) \Big]. \tag{3.10}$$

Summing rather than averaging over the 18 questions is what Cao et al. specify [9]. The choice matters: averaging would divide every gradient reaching the shared projection and the cut-points by 18, which at a fixed learning rate would shrink the distance those parameters can travel by the same factor. Averaging over the batch is the reduction used by all the other losses here, so the two conventions are combined as written in Equation 3.10.

3.6.7 Soft-QWK, and Why It Needs Two Phases

The metric used to judge every method, quadratic weighted kappa, cannot be optimised directly, because it is computed from counts of discrete predictions and counting has no useful gradient. The STBW system [46] replaces each discrete step by a smooth one, following earlier work on differentiable kappa objectives [16] and on turning a scalar prediction into a soft distribution [19]. This thesis implements the same objective.

First, the scalar prediction is clamped into the valid range and spread over the 19 levels with a Gaussian window of width $\sigma = 1$, normalised to sum to one:

$$\hat{y} = \mathrm{clip}(\hat{s}, 0, 18), \qquad P_i \propto \exp\left(-\frac{(i-\hat{y})^2}{2\sigma^2}\right). \tag{3.11}$$

The window is centred on the continuous prediction rather than on the rounded one, because a rounded centre would again have zero gradient. Second, with $\mathbf{T}$ the matrix of one-hot gold labels and $\mathbf{P}$ the matrix of soft predictions for the batch, a soft agreement matrix and a chance agreement matrix are accumulated,

$$O = \mathbf{T}^\top \mathbf{P}, \qquad E = \frac{\mathbf{o}_{\text{gold}}\, \mathbf{o}_{\text{pred}}^\top}{N}, \tag{3.12}$$

where $\mathbf{o}_{\text{gold}}$ and $\mathbf{o}_{\text{pred}}$ are the row and column totals of O. Third, with the quadratic penalty $W_{ij} = (i-j)^2$, the differentiable kappa and the loss are

$$\kappa_{\text{soft}} = 1 - \frac{\langle W, O\rangle}{\langle W, E\rangle + \varepsilon}, \qquad \mathcal{L}_{\text{Soft-QWK}} = 1 - \kappa_{\text{soft}}, \qquad \varepsilon = 10^{-9}. \tag{3.13}$$

This objective has a property no other loss in the comparison has: it is a statistic of the whole batch, not a sum of per-sentence terms. A model can therefore lower it by placing every prediction near the mean level of the batch, which keeps the soft agreement matrix plausible without reading any

sentence. The STBW authors report exactly this failure and avoid it by training in two phases: mean squared error first, then Soft-QWK from that checkpoint [46]. The same two-phase schedule is used here. Phase one trains with mean squared error for three epochs and keeps the checkpoint with the lowest development loss; phase two continues from that checkpoint with the Soft-QWK objective for three epochs, with a fresh optimiser and a fresh learning rate schedule. The total budget stays at six epochs, so the method is not given more training than the others, but its schedule is different and its results are therefore reported as a separate group throughout Chapter 4.

Three values of this method are not specified by the STBW paper and are choices made here: the Gaussian width $\sigma = 1$, the position of the phase boundary after epoch three, and the numerical guard ε. The phase boundary follows the STBW observation that their warm-up checkpoint was the third epoch.

3.6.8 Numerical Safeguards

Two losses take a logarithm of a quantity that a softmax can drive to a boundary. In OLL, a predicted probability of exactly one at some level would make $\log(1 - p_k)$ infinite, and at the gold level that infinity would be multiplied by a zero weight, producing an undefined value; the probability is therefore clamped just below one before the logarithm. In the soft-label loss the probability is clamped just above zero before $\log p_i$. Both clamps use the smallest representable margin of the working precision, so they agree with the literal formula wherever the literal formula is finite. All losses upcast the model output to single precision before any softmax, logarithm or sigmoid, so that mixed-precision training does not change the arithmetic of the loss itself.

3.6.9 Verification of the Loss Implementations

Each loss is checked against an independent implementation before any training run, on the same random logits in double precision. Cross-entropy, OLL, EMD and the four soft-label losses are compared against literal element-by-element loops written in the form used by the reference script; CORAL is compared against a closed-form numerically stable expansion of the binary cross-entropy; and Soft-QWK, evaluated with a very narrow Gaussian so that its soft distribution approaches a one-hot vector, is compared against $1 - \kappa$ computed by `scikit-learn` on hard predictions. Every check agrees to better than 10^{-9}, and the program refuses to train if any of them fails. The metric implementations are verified in the same way, as described in Section 3.11.

3.7 Loss Functions Added from Readability Research in Other Languages

The losses above are those of the BAREC benchmark and of the two shared-task systems closest to this work. This section adds three that none of them evaluates, as the part of the thesis that goes beyond the benchmark. They were chosen with three criteria: the loss has been used for readability assessment in another language, or a BAREC system paper names it as future work; its formula can be stated exactly from the authors' paper or code; and it has a matched control in the comparable group, so that the comparison changes one thing only. Each run uses the corpus, the splits, the input, the encoder, the optimiser, the learning rate, the schedule, the number of epochs, the seed and the checkpoint selection rule of Table 3.3.

3.7.1 CORN

CORN [44] keeps the 18 binary questions of CORAL, "is the level greater than k?", and changes two things. First, the questions no longer share a projection: the head is a linear layer with 18 outputs, $z_k = \mathbf{w}_k^\top \mathbf{h} + b_k$, each with its own weight vector. Second, question k is trained only on the sentences whose level is at least k, so that $\sigma(z_k)$ estimates the conditional probability $P(y > k \mid y \geq k)$. With $S_k = \{n : y^{(n)} \geq k\}$ the set of sentences of the batch that question k sees, and $t_k^{(n)} = \mathbb{1}[y^{(n)} > k]$, the loss is

$$\mathcal{L}_{\text{CORN}} = -\frac{1}{\sum_k |S_k|} \sum_{k=0}^{17} \sum_{n \in S_k} \Big[t_k^{(n)} \log \sigma\big(z_k^{(n)}\big) + \big(1 - t_k^{(n)}\big) \log \big(1 - \sigma\big(z_k^{(n)}\big)\big) \Big]. \tag{3.14}$$

The unconditional probabilities are recovered by the chain rule, $P(y > k) = \prod_{j \leq k} \sigma(z_j)$, which cannot increase with k, so the prediction is rank-consistent by construction without any constraint on the weights, and the level is the number of these probabilities above 0.5:

$$\hat{y} = \sum_{k=0}^{17} \mathbb{1}\Big[\prod_{j \leq k} \sigma(z_j) > 0.5 \Big]. \tag{3.15}$$

The ZAI system paper names CORN as the next ordinal objective to try on BAREC [40], and the CORAL run of the comparable group is its matched control: both use 18 binary questions on the same pooled representation, and they differ in the shared projection and in the conditional training subsets.

3.7.2 Pairwise Ranking Losses

A second line of readability research treats difficulty as a ranking: a model should score a harder text above an easier one [31, 47]. A ranking score alone has no scale, so it cannot be decoded into 19 levels. The two ranking objectives are therefore added to the regression loss, on the regression head,

and decoded exactly as the regression method is. With $\hat{s}_n$ the output for sentence n and $\mathcal{P} = \{(i, j) : y^{(i)} > y^{(j)}\}$ the ordered pairs of the batch whose gold levels differ,

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \left(\hat{s}_n - y^{(n)}\right)^2 + \lambda \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \ell\left(\hat{s}_i - \hat{s}_j\right), \qquad \lambda = 1, \tag{3.16}$$

where the pair loss ℓ is one of the two below. Pairs of equal level carry no order and do not enter; a batch with a single level adds nothing to the regression loss.

Margin ranking (RANK). The margin ranking loss of the multilingual Wikipedia readability model of Trokhymovych et al. [47],

$$\ell_{\mathrm{RANK}}(x) = \max(0,\, m - x), \qquad m = 0.5, \tag{3.17}$$

penalises a pair whose harder sentence is not scored at least m above the easier one.

Pairwise logistic (PLR). The pairwise logistic loss, the RankNet loss [8], is the one the neural pairwise ranking model of Lee and Vajjala is trained with: their released code compiles the model with the `PairwiseLogisticLoss` of TF-Ranking [31]. It is

$$\ell_{\mathrm{PLR}}(x) = \log\left(1 + e^{-x}\right), \tag{3.18}$$

computed in the numerically stable form $\max(0, -x) + \log(1 + e^{-|x|})$ used by TF-Ranking. It never reaches zero, so every correctly ordered pair still pushes the two scores further apart, while the hinge of RANK stops at the margin.

The regression run of the comparable group is the matched control of both, and RANK and PLR differ from each other only in ℓ. Two adaptations

are ours and are declared here. Both papers form pairs from versions of the same article at different levels, which BAREC does not have, so the pairs are formed between the sentences of a training batch. And the weight $\lambda = 1$, which adds the two terms without tuning, appears in neither paper.

3.7.3 Batch Composition for the Added Losses

All three added losses are statistics of the batch rather than sums over sentences: CORN divides by the number of sentences in the conditional subsets of the batch, and the ranking terms average over the pairs inside it. Gradient accumulation would compute them on 16 sentences at a time instead of 64, so, like the second phase of Soft-QWK, they are trained with a true batch of 64 sentences and gradient checkpointing (Section 3.9). The number of optimiser steps per epoch is unchanged. The two ranking losses are checked before training against a double loop over the pairs and against the PyTorch implementation of the margin ranking loss and of the softplus function, to better than 10^{-9}.

3.8 Cut-Point Initialisation for CORAL

The 18 cut-points of the CORAL head are ordinary parameters, and the natural default is to start them at zero. Under the BAREC protocol that default is a problem, and the reason is arithmetic rather than statistical.

At convergence, cut-point b_k has to sit near the value at which the shared score s separates level k from level $k+1$. Since the 19 levels are very unevenly populated, those values are spread over a wide interval: the fitted spread between the highest and the lowest cut-point in this work is 11.68 (Chapter 4). Now consider how far a parameter can travel under the protocol. Training runs for six epochs of 857 optimiser steps, that is 5,142 steps, at a learning rate of 5×10^{-5} decayed linearly to zero. The sum of the learning rate over all steps is 0.13. With an optimiser whose per-step update is bounded in

magnitude by the learning rate, as AdamW's normalised update effectively is [35], no single parameter can move much further than 0.13 in total. A cut-point that starts at zero cannot reach a value near -6 or $+5$, no matter how clear the training signal is.

The initialisation used here removes the problem instead of spending the budget on it. The cumulative label frequencies of the *training* split are converted into the cut-point that reproduces them before any sentence is seen:

$$b_k = \log \frac{\hat{P}(y > k)}{1 - \hat{P}(y > k)}, \qquad \hat{P}(y > k) = \frac{1}{N_{\text{train}}} \sum_{n=1}^{N_{\text{train}}} \mathbb{1}\left[y^{(n)} > k\right], \tag{3.19}$$

with the frequency clipped to $[10^{-4}, 1 - 10^{-4}]$ so that the logit stays finite. Since $\hat{P}(y > k)$ is non-increasing in k by construction, the initial cut-points are ordered, which the implementation also asserts. At initialisation the head therefore predicts the training label distribution with a score of zero, and the training budget is spent on learning the projection that separates sentences rather than on transporting biases across the real line. Only training labels are used, so the development and test splits are untouched by the initialisation.

This is a head initialisation, not a change of the pipeline, and it is the one element of the method that is neither in the BAREC script nor in the two shared-task papers. Two experiments in Chapter 4 test the argument rather than assume it: the distance the cut-points actually travel during training is measured, and an ablation raises the head learning rate to 10^{-3}, which multiplies the budget by twenty while leaving the encoder at the protocol value.

3.9 Training Protocol

3.9.1 Hyper-parameters

Table 3.3 lists the protocol. Every value is the one used by the BAREC script, including those that the script leaves at the library default: no warm-up, no weight decay, a linear decay schedule and gradient clipping at a norm of one.

Table 3.3: Training protocol, identical for every method in the fine-tuning track.

Setting	**Value**
Encoder	`aubmindlab/bert-base-arabertv2`
Input	D3Tok column of the corpus
Maximum sequence length	512 subword tokens, dynamic padding
Optimiser	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
Learning rate	5×10^{-5}
Schedule	linear decay to zero, no warm-up
Weight decay	0
Gradient clipping	maximum norm 1.0
Epochs	6
Effective training batch	64 sentences (16 per step, 4 accumulation steps)
Evaluation batch	16
Optimiser steps per epoch	857
Mixed precision	fp16 autocast, single-precision master weights
Random seed	42
Checkpoint selection	lowest development loss over the six epochs, no early stopping

3.9.2 Checkpoint Selection

The benchmark selects the epoch with the lowest development loss, keeps no other criterion and uses no early stopping; that rule is used here for every method, including the ones whose source papers use a different rule. This matters for the comparison, because the development loss of each method is measured in that method's own loss, so the rule is not a metric shared between methods but a per-method criterion. The two shared-task systems both select on development QWK instead [40, 46], which raises the question of how much of their advantage comes from the selection rule. Section 3.10 therefore declares a selection ablation that evaluates the highest-QWK epoch of every method as well.

Reporting the numbers of the selected epoch requires care. The selected model is saved explicitly at the moment it is identified, the saved weights are reloaded for evaluation, and the reloaded development metrics are compared with what was logged during training at that same epoch. The comparison is recorded in every result file and is reported in Chapter 4; it is the check that the reported numbers belong to the selected epoch and not to some later one.

3.9.3 Differences from the Published Script

Three implementation differences are declared here because they are visible in the arithmetic, and one because it is a correction. None of them changes a hyper-parameter, and all of them are identical across methods.

1. **Batch composition.** A batch of 64 sentences at length 512 does not fit in the 16 gigabytes of the available accelerator, so the batch is formed as 16 sentences per forward pass with 4 accumulation steps. The 54,845 training sentences then give 857 optimiser steps per epoch, the same number as a true batch of 64. Each optimiser step therefore averages the gradient of 64 sentences, with one exception: the last step of each

epoch covers the remaining 61 sentences as four micro-batches of 16, 16, 16 and 13 sentences, so its per-sentence weighting differs slightly from the weighting a single batch of 61 sentences would produce.

2. **Mixed precision.** Training uses fp16 autocast with single-precision master weights, which the reference script does not; it was adopted to fit the available time budget. Every loss is computed in single precision, as described above. Mixed precision can cause the optimiser to skip a step whose gradients overflow, and the number of such skipped steps was not recorded, so it cannot be ruled out that a small number of updates were dropped.

3. **Batch composition for Soft-QWK and the added losses.** Gradient accumulation is equivalent to a larger batch only for losses that are sums over sentences. Soft-QWK, CORN and the two ranking losses are statistics of the whole batch, so accumulation would compute them on 16 sentences at a time instead of 64. The second phase of Soft-QWK and the three added losses are therefore trained with a true batch of 64 sentences and gradient checkpointing, which trades computation for memory without changing the arithmetic.

4. **Methods the script does not implement.** The published script contains no regression trainer, and it routes the name CORAL to an ordinary classification trainer. The regression method here follows the description in the benchmark paper, and CORAL follows Cao et al. [9]. The weighted kappa loss present in the script is not run, on the supervisor's instruction, since it appears in no table of the benchmark paper and is neither CORAL nor Soft-QWK.

3.10 Methods Compared and Ablations

Table 3.4 lists the fourteen training runs. Nine form the comparable group, in which the protocol is identical and only the head and the loss change. Soft-QWK is reported as its own group because its schedule differs, as explained above. The ablation answers a specific question and is never mixed into the comparable group. The three added losses of Section 3.7 follow the comparable protocol and are reported beside their matched controls.

Table 3.4: The fourteen runs of the fine-tuning track.

Run	**Head**	**Group**	**Role**
CE	classification	comparable	ordinal-blind baseline
Regression (MSE)	regression	comparable	best benchmark loss
OLL, $\alpha = 1.5$	classification	comparable	distance-weighted loss
EMD	classification	comparable	cumulative distribution loss
SOFT, $\beta = 2, 3, 4, 5$	classification	comparable	distance-weighted targets
CORAL	CORAL	comparable	ordinal head, the ZAI approach
MSE $\rightarrow$ Soft-QWK	regression	two-phase	metric-aligned, the STBW approach
CORAL, head lr 10^{-3}	CORAL	ablation	tests the cut-point travel argument
CORN	CORN	added	successor of CORAL; control CORAL
MSE + margin ranking (RANK)	regression	added	multilingual readability ranking; control MSE
MSE + pairwise logistic (PLR)	regression	added	neural pairwise ranking; control MSE

Two further ablations need no additional training, because they re-evaluate saved models.

Decision threshold for CORAL. The ZAI system tunes a single decision threshold on the development set instead of using 0.5 [40]. The saved CORAL model is therefore decoded again with every threshold on the grid $0.05, 0.06, \ldots, 0.95$, the value with the highest development QWK is kept, and the test numbers under that threshold are reported. Its development numbers are optimistic by construction, since the threshold was chosen on that split.

Selection rule. As announced above, the highest development QWK epoch of every method is also saved and evaluated. Here too the development numbers are optimistic by construction and only the test column can be read as a result.

3.11 Evaluation Metrics

Six metrics come from the benchmark, and three more are added.

- **QWK**, quadratic weighted kappa [11], computed with `scikit-learn` with quadratic weights. It is the primary metric of the benchmark and of the shared task, and it is the metric on which all comparisons in this thesis are based. It rewards agreement while penalising an error by the square of its size, which is exactly the ordinal property the losses try to exploit.
- **Acc19**, the proportion of exactly correct levels.
- **±1 Acc19**, the proportion of predictions within one level of the gold level.
- **Dist**, the mean absolute distance between predicted and gold level, in levels.

- **Acc7, Acc5, Acc3**, exact accuracy after collapsing predictions and gold labels to the coarser 7-, 5- and 3-level schemes with the corpus's own mappings, verified as described in Section 3.3.
- **far≥3** and **far≥5**, the proportion of predictions that are at least three and at least five levels away from the gold level. These two are *not* BAREC metrics; they are introduced in this thesis to answer the part of the research brief that asks for the error distance, and specifically to separate a near miss from a far miss. ±1 accuracy counts the near misses and these two count the tails. They are labelled as this thesis's own wherever they appear.
- **Macro-F1** over the 19 levels, reported as a diagnostic of whether a method uses the whole label range or concentrates on the frequent levels.

The metric code is tested before use: a perfect predictor must give QWK, Acc19 and all collapsed accuracies of one and a distance of zero; a predictor shifted by exactly one level must give an Acc19 of zero, a ±1 accuracy of one and a distance of exactly one; and distance, ±1 accuracy and far≥3 are recomputed directly from the absolute differences and compared with the library results. The three collapse tables are also checked to be onto the 7, 5 and 3 target levels.

3.12 Frozen-Backbone Track

The research brief asks for CPU-friendly training settings. Fine-tuning 135 million parameters does not meet that description on any processor, so the requirement is met by a second track in which the encoder is not trained at all.

The encoder is run once over the corpus in inference mode and two sentence representations are cached for every sentence: the pooled output, as

in the fine-tuning track, and the mean of the final-layer token representations over the non-padding tokens. Only a linear head is then trained on the cached features, which takes a few minutes per run on a laptop processor; the 5 runs of every method together take 125 minutes.

The head hyper-parameters cannot be the protocol's. A linear head on frozen features, trained for six epochs at 5×10^{-5}, does not converge: the travel budget argument of Section 3.8 applies to every head parameter, not only to CORAL's cut-points. The head is therefore trained for 50 epochs at a learning rate of 10^{-3}, which gives 42,850 steps and a budget of 21.4, with the other settings unchanged: AdamW, no weight decay, linear decay, gradient clipping at one, dropout 0.1 and a batch of 64. These values are ours, they are listed here because they are not the benchmark's, and they are identical for every method in the track, which is what the comparison requires. Every method is run with five seeds, 42 to 46, and reported as a mean with a standard deviation. The checkpoint selection rule, the losses, the decoders, the collapse tables and the metrics are the same code as in the fine-tuning track.

This track also carries one ablation that the fine-tuning track cannot afford: CORAL with zero-initialised cut-points. Here the head budget is 21.4, far more than the spread the cut-points need, so the initialisation of Section 3.8 should make little difference; under the fine-tuning protocol the budget is 0.13 and it should make a large one. The two tracks together therefore test the argument from both sides.

Because the encoder never adapts, the absolute numbers of this track are much lower than the fine-tuning numbers, which is the expected outcome and not a defect. They are never compared with the published tables.

3.13 Implementation and Reproducibility

The fine-tuning track is a single self-contained script, run in three parts on three accounts of a free notebook service so that the eleven runs could proceed in parallel; the part is only a scheduling label and no hyper-parameter depends on it. Each run took between 25 and 30 minutes on one NVIDIA Tesla T4, and the eleven runs together took 291 minutes of accelerator time, that is about 4.8 hours. The three added losses were run afterwards as two further parts of the same script, whose only changes are the added loss functions, their tests and their runs; together they took 106 minutes on the same accelerator type, with the same library versions (Transformers 5.0.0) and the same corpus revision.

Every result file records the software and hardware it was produced on: Python 3.12.13, PyTorch 2.10.0+cu128, Transformers 5.0.0, Datasets 5.0.0, the accelerator name, the number of visible accelerators, the realised training batch, the observed number of optimiser steps per epoch, whether mixed precision was active, the corpus repository revision (3a33652778) and the encoder repository revision (97522efce1). Pinning the two repository revisions matters because both are living repositories that can change after publication.

Three guards run in every training run, and each aborts the run rather than produce a number that cannot be trusted. The first refuses to start if more than one accelerator is visible or if the realised training batch is not 64; an earlier version of these experiments silently trained at a batch of 128 because a second accelerator was present, and this guard exists to make that failure impossible. The second records the number of optimiser steps actually performed in each epoch and checks it against the number implied by the corpus size, the micro-batch and the accumulation. The third is the reloaded-checkpoint comparison of Section 3.9. The outcome of all three is reported for every method in Chapter 4.

3.14 Summary

This chapter fixed everything except the head and the loss: the corpus and its official splits, the D3Tok input read from the corpus itself, the AraBERTv2 encoder, and a training protocol taken from the benchmark script down to the library defaults. Nine runs form the comparable group, one two-phase run implements the metric-aligned objective, and two ablations test the cut-point argument and the decision threshold, with a third ablation testing the checkpoint selection rule. Three further runs add losses from readability research in other languages, CORN and two pairwise ranking objectives, each beside a matched control. Every loss and every metric was verified against an independent implementation before training, the three differences from the published script were declared, and the cut-point initialisation of Equation 3.19 was introduced with an explicit prediction about what the experiments should show. A second, frozen-backbone track provides the CPU-friendly setting of the research brief and the variance estimates that a single-seed protocol cannot give.

The next chapter reports what these fourteen runs produced, checks them against the published benchmark, and tests the predictions made here.

Chapter 4

Experiments and Results

4.1 Introduction

This chapter reports the eleven training runs described in Chapter 3 and what they say about the research question. Section 4.2 reports the checks that decide whether the numbers can be trusted at all. Section 4.3 compares the eight loss functions that the BAREC benchmark also reports, which is the evidence that the pipeline is faithful. Section 4.4 presents the full results on both splits and answers the question put by the research brief: how CORAL and the QWK-oriented objective compare with the distance-aware losses under one identical protocol. Sections 4.5 and 4.6 separate two things that the single headline metric hides, namely exact-match accuracy and the distribution of error distances. Sections 4.7 to 4.9 report the three ablations of the fine-tuning track, Section 4.10 the two-phase Soft-QWK method, Section 4.11 the three losses added from readability research in other languages, and Section 4.12 the frozen-backbone track with its own ablation. Section 4.13 places the numbers beside the published systems and beside human agreement, and Section 4.14 states the limitations.

Every number in this chapter is read from the result files produced by the training runs. The values quoted in the text and the values in the tables

are generated from those files by a script, so the text and the tables cannot disagree.

4.2 Integrity of the Runs

The three guards declared in Section 3.13 passed identically in all eleven runs, and the result files record the evidence.

- Exactly one accelerator was visible in every run, an NVIDIA Tesla T4, and the realised training batch was 64 sentences in every run, including both phases of the two-phase method.
- Every epoch of every run performed 857 optimiser steps, which is the number implied by 54,845 training sentences at an effective batch of 64.
- The reloaded selected checkpoint reproduced the development QWK, accuracy and distance logged during training at that epoch, to within 10^{-9}, in every run. The numbers reported below therefore belong to the epoch the selection rule chose.
- All eleven runs used the same corpus revision (3a33652778) and the same encoder revision (97522efce1), with Python 3.12.13, PyTorch 2.10.0+cu128 and Transformers 5.0.0, and none of them was a reduced smoke run.

The eleven runs together took 291 minutes of accelerator time, between 25 and 30 minutes each.

4.3 Reproduction of the Benchmark

Eight of the nine methods in the comparable group are loss functions that the BAREC benchmark also reports, on the same corpus, the same encoder

and the same input representation. Comparing them is therefore a test of the pipeline rather than a contribution, and it has to be passed before anything else in this chapter can be believed. Table 4.1 gives the comparison on the development set.

Table 4.1: Reproduction of the eight loss functions of BAREC Table 6 on the development set. Paper values are from Elmadani et al. [24]; the gap is ours minus theirs.

	QWK		**Acc19**		**±1 Acc19**		**Dist**	
Loss	ours	paper	ours	paper	ours	paper	ours	paper
CE	79.81	80.0	54.84	56.6	70.47	69.9	1.163	1.14
Regression (MSE)	82.88	84.0	35.54	43.1	71.15	73.1	1.212	1.13
OLL ($\alpha = 1.5$)	82.91	82.8	46.21	47.3	71.15	71.1	1.130	1.13
EMD	81.29	81.2	54.27	55.3	70.10	70.3	1.125	1.11
SOFT ($\beta = 2$)	81.41	80.0	57.07	55.8	71.66	69.8	1.088	1.15
SOFT ($\beta = 3$)	79.72	80.1	55.40	56.4	69.44	69.9	1.177	1.14
SOFT ($\beta = 4$)	80.05	79.6	55.62	56.4	70.79	69.9	1.144	1.15
SOFT ($\beta = 5$)	80.63	79.3	55.85	56.2	70.86	69.5	1.134	1.17

Seven of the eight losses land within 1.5 QWK points of the published value, and the mean absolute difference over the eight is 0.64 points. Cross-entropy reproduces to -0.19 points, EMD to +0.09 and OLL to +0.11, which is as close as a single run of a fine-tuning pipeline can reasonably be expected to land. The soft-label family sits above the published values, most clearly at $\beta = 2$ (+1.41) and $\beta = 5$ (+1.33), and its ordering across β is not the published one. The four values span 1.69 QWK points in our run while the published four span 0.8, so this is the family in which a single run is least informative.

The one substantial difference is the regression method, at -1.12 QWK points and -7.56 points of exact accuracy. Table 4.2 explains most of it.

Table 4.2: Development QWK after each epoch. The selected epoch, that is the epoch with the lowest development loss, is starred. The two-phase method is shown as its three mean squared error epochs followed by its three Soft-QWK epochs.

Method	**ep. 1**	**2**	**3**	**4**	**5**	**6**
CE	79.12	**79.81***	81.12	80.40	80.83	80.45
Regression (MSE)	81.90	**82.88***	83.51	83.55	82.83	83.08
OLL ($\alpha = 1.5$)	81.71	82.57	**82.91***	82.33	82.41	82.43
EMD	79.63	**81.29***	81.07	81.08	80.80	81.28
SOFT ($\beta = 2$)	79.96	**81.41***	81.58	81.57	80.69	80.70
SOFT ($\beta = 3$)	**79.72***	80.69	80.89	80.71	80.15	80.46
SOFT ($\beta = 4$)	78.74	**80.05***	81.09	80.77	80.87	80.45
SOFT ($\beta = 5$)	79.73	**80.63***	81.18	81.12	80.86	80.63
CORAL	81.23	**82.90***	83.11	82.32	82.07	82.09
CORAL, head lr 10^{-3}	81.93	83.41	**83.86***	82.79	82.41	82.42
MSE warm-up (phase 1)	81.24	82.54	83.10		–	
Soft-QWK (phase 2)		–		82.13	**83.24***	82.57

The development QWK of the regression method keeps rising until epoch four, but its development *loss* is lowest at epoch 2, and the selection rule takes that epoch. At epoch four the method reaches 83.55 development QWK and 42.08 exact accuracy, against the published 84.0 and 43.1; on the test set it reaches 84.32 QWK and 41.22 exact accuracy against the published 84.4 and 41.4. In other words the implementation does reach the published numbers, and the gap in Table 4.1 is the price of one checkpoint decision, taken by a rule that both the benchmark and this thesis apply. The mean squared error loss is only loosely coupled to QWK, so the epoch that minimises it need not be the epoch that maximises the metric; the same effect appears, smaller, for several other methods and is quantified in Section 4.9.

Table 4.3 repeats the comparison at the coarser granularities that the benchmark reports for three of the losses.

Table 4.3: Collapsed granularities for the three losses the BAREC paper reports at 7, 5 and 3 levels (Tables 7 and 8 of Elmadani et al. [24]).

		Acc7		**Acc5**		**Acc3**	
Split	**Loss**	ours	paper	ours	paper	ours	paper
dev	CE	65.16	65.9	69.96	70.3	76.76	76.5
dev	EMD	65.01	65.2	70.10	70.0	76.63	76.4
dev	Regression (MSE)	57.02	61.1	64.43	67.8	73.34	75.9
test	CE	64.36	65.1	68.71	69.4	74.81	75.2
test	EMD	64.00	64.9	68.97	69.0	74.87	75.2
test	Regression (MSE)	56.24	59.4	63.05	65.3	71.31	72.8

Cross-entropy and EMD agree with the published collapsed accuracies to within about one point on both splits. The regression method is again lower, by three to four points, which is consistent with the same checkpoint effect: collapsed accuracy is an exact-match measure, and the epoch chosen for the regression method is the one at which its predictions are still concentrated near the centre of the scale.

The conclusion of this section is that the pipeline reproduces the benchmark. That licence is what allows the next sections to add CORAL and Soft-QWK to the same table and to read the differences as differences between methods.

4.4 Main Results

Tables 4.4 and 4.5 report every metric for every run on the development and test splits.

Table 4.4: All methods on the BAREC development set. AraBERTv2 with D3Tok input, one seed.

Method	**QWK**	**Acc19**	**±1**	**Dist**	**Acc7**	**Acc5**	**Acc3**	**far≥3**	**far≥5**	**mF1**	**Ep.**
Comparable group: identical protocol, one head and one loss changed											
CE	79.81	54.84	70.47	1.163	65.16	69.96	76.76	15.60	6.69	45.45	2
Regression (MSE)	82.88	35.54	71.15	1.212	57.02	64.43	73.34	13.43	3.67	22.13	2
OLL ($\alpha = 1.5$)	82.91	46.21	71.15	1.130	61.55	67.62	76.33	13.45	4.24	27.64	3
EMD	81.29	54.27	70.10	1.125	65.01	70.10	76.63	14.12	5.84	36.60	2
SOFT ($\beta = 2$)	81.41	57.07	71.66	1.088	66.95	71.23	77.44	14.23	5.84	46.88	2
SOFT ($\beta = 3$)	79.72	55.40	69.44	1.177	65.20	69.59	76.01	15.25	6.92	40.05	1
SOFT ($\beta = 4$)	80.05	55.62	70.79	1.144	65.94	70.21	76.84	15.17	6.31	43.88	2
SOFT ($\beta = 5$)	80.63	55.85	70.86	1.134	66.21	70.26	77.02	15.23	6.32	44.79	2
CORAL	82.90	34.15	66.84	1.289	55.40	63.60	73.00	12.86	4.32	24.56	2
Two-phase group: MSE warm-up then Soft-QWK, so the schedule differs											
MSE → Soft-QWK	83.24	35.23	72.33	1.251	58.29	65.76	75.31	13.83	4.73	28.03	2
Ablation: not part of the comparable group											
CORAL, head lr 10^{-3}	83.86	42.56	69.37	1.167	59.02	66.25	75.25	12.45	4.13	28.88	3

QWK, the accuracies, far≥3, far≥5 and macro-F1 are percentages; Dist is in levels; Ep. is the selected epoch. far≥3 and far≥5 are this thesis's own metrics.

Table 4.5: All methods on the BAREC test set. Same checkpoints as Table 4.4.

Method	QWK	Acc19	±1	Dist	Acc7	Acc5	Acc3	far≥3	far≥5	mF1	Ep.
Comparable group: identical protocol, one head and one loss changed											
CE	80.48	54.72	70.29	1.128	64.36	68.71	74.81	15.06	6.30	46.86	2
Regression (MSE)	83.64	34.20	71.70	1.186	56.24	63.05	71.31	13.41	2.88	23.61	2
OLL ($\alpha = 1.5$)	84.08	46.28	72.08	1.074	60.69	66.09	73.80	12.52	3.60	29.01	3
EMD	82.70	53.58	69.39	1.080	64.00	68.97	74.87	13.33	5.23	38.18	2
SOFT ($\beta = 2$)	81.86	55.96	71.18	1.070	65.43	69.60	75.60	14.00	5.56	47.19	2
SOFT ($\beta = 3$)	80.31	54.64	69.00	1.142	63.88	68.24	74.70	14.69	6.49	42.14	1
SOFT ($\beta = 4$)	80.94	55.61	70.93	1.095	65.18	69.39	75.20	14.40	5.82	46.04	2
SOFT ($\beta = 5$)	81.37	56.03	70.93	1.086	65.41	69.60	75.50	14.52	5.87	46.37	2
CORAL	84.33	34.15	66.70	1.232	54.09	60.98	69.21	12.06	3.29	26.38	2
Two-phase group: MSE warm-up then Soft-QWK, so the schedule differs											
MSE → Soft-QWK	84.31	35.96	72.21	1.196	57.33	64.36	72.80	13.07	4.19	28.96	2
Ablation: not part of the comparable group											
CORAL, head lr 10^{-3}	84.58	42.15	70.37	1.121	58.39	64.59	73.03	11.57	3.49	28.70	3

QWK, the accuracies, far≥3, far≥5 and macro-F1 are percentages; Dist is in levels; Ep. is the selected epoch. far≥3 and far≥5 are this thesis's own metrics.

4.4.1 CORAL and the QWK-Oriented Objective

The question in the research brief is whether an ordinal head and a metric-aligned loss beat distance-aware losses when the dataset, the protocol and the metric are held fixed. The answer is that they are at the top of the table, and that they are not alone there.

On the development set the comparable group is led by OLL at 82.91, CORAL at 82.90 and the regression method at 82.88. The distance between these three is 0.03 QWK points. The two-phase Soft-QWK method reaches 83.24, which is +0.70 above its own warm-up checkpoint and the highest development value outside the ablation, but it is a different training schedule and is therefore reported separately. On the test set the order of the leaders changes: CORAL reaches 84.33, Soft-QWK 84.31, OLL 84.08 and the regression method 83.64.

Two things follow, and they should be stated with the same care.

First, ordinal-aware objectives clearly beat the ordinal-blind baseline. Cross-entropy is 79.81 on development and 80.48 on test, which together with SOFT at $\beta = 3$ forms the bottom of the comparable group. Every other method is above it, by up to three QWK points on development and up to four on test; the single soft-label configuration that does not beat it is $\beta = 3$, at 79.72 and 80.31. This is the same conclusion the benchmark draws, and it is reproduced here under a protocol in which nothing but the head and the loss changed.

Second, the differences *among* the ordinal-aware objectives are small, and the ranking is not stable between splits: the leader on development is fourth on test, and CORAL rises from 82.90 on development to 84.33 on test while the regression method rises by less than a point. With one seed per method, differences of a few tenths of a QWK point cannot be attributed to the mechanism. This thesis therefore does not name a single best loss. What the evidence supports is a group of four objectives, namely the CORAL head, the metric-aligned two-phase schedule, distance-weighted ordinal log loss and plain regression, that are equivalent within the resolution of this experiment and that all clearly beat cross-entropy and the soft-label family on QWK.

4.4.2 Where the Groups Differ

Reading across Table 4.4 shows that the four leaders on QWK are not the same methods that lead on the other metrics, and that the split is systematic. The two methods with a continuous or cumulative output, that is the regression head and the CORAL head, have the highest QWK and the lowest exact accuracy: 35.54 and 34.15 respectively, against 57.07 for the best soft-label configuration. Their macro-F1 values, 22.13 and 24.56 against 46.88, say the same thing in a different way: they concentrate their predictions on a narrower band of levels and rarely commit to a rare level. The classification head with a distance-aware loss does the opposite: it spreads predictions across the whole scale, gets more of them exactly right, and pays for the

attempts with a longer tail of errors. Section 4.5 quantifies this trade-off and Section 4.6 looks at the tail itself.

4.5 Exact Match against Ordinal Agreement

The trade-off is visible as an almost exact inversion between two columns of the same table. Among the ten runs of the comparable and two-phase groups, the four highest development QWK values belong to Soft-QWK, OLL, CORAL and regression, and those same four hold the four lowest exact accuracies: 34.15 for CORAL, 35.23 for Soft-QWK, 35.54 for regression and 46.21 for OLL. Conversely the six methods with the highest exact accuracy, led by SOFT at $\beta = 2$ with 57.07, occupy the lower half of the QWK column.

The mechanism is not mysterious. A regression or cumulative model is rewarded for placing its prediction close to the gold level and is never rewarded for landing on it exactly, so it learns a smooth, cautious mapping and keeps away from the ends of the scale. A softmax model with a distance-aware loss is still trained to put mass on the gold level, so it keeps committing to specific levels, wins exact matches, and occasionally commits badly.

This has a practical consequence that the single headline metric hides. If the intended use is to place a text in a narrow band, for example to choose which grade a text suits, then QWK and the distance metrics are the right criteria and the regression or CORAL family is the right choice. If the intended use needs the exact level, for instance to label material by a curriculum level without human review, then exact accuracy matters and the soft-label family is nearly twenty points better at it than the QWK leaders, at a cost of two to three QWK points. Reporting only QWK, as both the benchmark and the shared task do, obscures a difference this large.

4.6 Error Distance: Near Misses and Far Misses

The research brief asks not only which method scores best but which produces the smallest error distance, and specifically whether errors are near misses rather than far misses. Four columns of Tables 4.4 and 4.5 answer that: adjacent accuracy, mean distance, and the two far-miss rates introduced in this thesis.

The first observation is that the best method on QWK is not the best method on distance. On the development set the lowest mean distance in the study is 1.088 levels, from SOFT at $\beta = 2$, whose QWK is 81.41, two points below the leaders. The QWK leaders have *worse* mean distance: 1.289 for CORAL and 1.251 for Soft-QWK, the two worst values in the whole table. The same pattern holds on test, where the lowest mean distance is 1.070 and CORAL is at 1.232.

This looks contradictory only until the two tails are separated. Mean distance averages all errors, so it is dominated by the many small ones; QWK weights an error by its square, so it is dominated by the few large ones. The far-miss rates show which methods control the large errors. On the development set the lowest rate of errors of three levels or more is 12.86 per cent, from CORAL, followed by the regression method at 13.43 and OLL at 13.45; cross-entropy is worst at 15.60. For errors of five levels or more the order is the regression method at 3.67 per cent, OLL at 4.24 and CORAL at 4.32, against 6.69 for cross-entropy. On the test set CORAL again has the fewest errors of three levels or more, 12.06 per cent, and the regression method the fewest of five levels or more, 2.88.

So the ordinal mechanisms do exactly what they were designed to do, and the effect is specific: cross-entropy makes roughly one error of five levels or more in every fifteen sentences, while the regression head makes one in every twenty-seven on the development set. What the ordinal mechanisms do not

do is reduce the number of small errors; the soft-label family, which keeps a cross-entropy-like target, remains better at that.

Adjacent accuracy behaves like a third, intermediate criterion. The highest development value is 72.33 per cent for Soft-QWK, followed by 71.66 for SOFT at $\beta = 2$ and 71.15 for regression, while CORAL is last at 66.84. CORAL therefore achieves its high QWK without being good at landing within one level: it avoids the catastrophic errors that QWK punishes hardest, but it also misses the adjacent band more often than any other method. For an application in which a one-level error is acceptable and a three-level error is not, CORAL and the regression method are the appropriate choices; if a one-level error already has a cost, the soft-label family is preferable.

Taken together, the three views of error distance answer the brief's question in a way a single metric cannot: the methods that maximise QWK are the methods that minimise far misses, not the methods that minimise average error.

4.7 CORAL Cut-Points and the Travel Budget

Section 3.8 argued that CORAL's cut-points cannot travel far under the BAREC protocol, and initialised them from the training label frequencies for that reason. Two measurements test the argument.

The first is the distance the cut-points actually moved. Over the whole run, no cut-point moved further than 0.0703 from its initial value, against a fitted spread of 11.68 between the highest and the lowest cut-point. The budget computed in Section 3.8 was 0.13, and the observed movement is well inside it. The trained cut-points run from 5.16 down to -6.52, which is to say they are, to two decimal places, the values that Equation 3.19 wrote into them before training started. Under this protocol the cut-points are not learned; they are supplied. A zero-initialised head would have had to move each of

them by up to six units on a budget of 0.13, which the arithmetic forbids. Table 4.6 also confirms that the 18 cut-points stayed correctly ordered after every epoch, so the rank consistency that motivates the CORAL head held throughout training and not only at the optimum.

Table 4.6: CORAL cut-point spread $b_0 - b_{17}$ after each epoch, and whether the 18 cut-points were still ordered. Measured on the saved model at the end of every epoch; the largest distance any cut-point ever travelled from its initial value is 0.0703 with the shared learning rate and 1.59 in the head learning rate ablation.

Method	**ep. 1**	**2**	**3**	**4**	**5**	**6**	**ordered**
CORAL	11.62	11.68	11.73	11.76	11.78	11.79	yes
CORAL, head lr 10^{-3}	12.65	13.62	14.43	15.01	15.38	15.51	yes

The second measurement is the ablation that raises the head learning rate to 10^{-3} and leaves the encoder at 5×10^{-5}, multiplying the cut-point budget by twenty to 2.57. The cut-points then move, by up to 1.59, and their spread widens from 11.68 to 14.43. Development QWK rises by +0.96 points to 83.86 and test QWK by +0.25 points to 84.58, the highest test value recorded in this work. Exact accuracy rises as well, from 34.15 to 42.56, and the far-miss rate falls from 12.86 to 12.45.

Both measurements are consistent with the argument, and the ablation adds something the argument did not predict: the cut-points are not merely hard to move, they are worth moving. Since the ablation changes an optimiser setting rather than the head, it is reported outside the comparable group; it is also a single run, so the size of its gain should be treated as an indication and not as a measured effect. The honest summary is that the initialisation of Equation 3.19 makes a CORAL head usable under a protocol whose learning rate was chosen for an encoder, and that a head-specific learning rate appears to be the better remedy.

4.8 Decision Threshold for CORAL

The ZAI system decodes its CORAL head with a threshold tuned on the development set rather than with 0.5 [40]. Repeating that on the saved CORAL model selects $t = 0.51$, barely away from the default, and moves test QWK by +0.12 points, from 84.33 to 84.45. In the head learning rate ablation the grid selects $t = 0.50$ and nothing changes.

The threshold is therefore not where the ZAI system's advantage comes from, at least not with cut-points initialised as they are here. This is what Equation 3.19 predicts: an initialisation that already encodes the cumulative label frequencies places the decision boundaries where the data puts them, so there is little left for a global threshold shift to correct.

4.9 The Checkpoint Selection Rule

Both shared-task systems select their checkpoint on development QWK [40, 46], while the benchmark, and therefore this thesis, selects on development loss. Table 4.7 evaluates both rules for every method. Only the test columns can be read as results, since a checkpoint chosen on development QWK has an optimistic development QWK by construction.

Table 4.7: Checkpoint selection ablation. The left block is BAREC's rule (lowest development loss), the right block the rule used by the two shared-task systems (highest development QWK). Development QWK under the right rule is optimistic by construction, so only the test column should be read as a result.

	lowest dev loss			**highest dev QWK**			
Method	epoch	dev QWK	test QWK	epoch	dev QWK	test QWK	**Δ test**
CE	2	79.81	80.48	3	81.12	81.80	+1.32
Regression (MSE)	2	82.88	83.64	4	83.55	84.32	+0.67
OLL ($\alpha = 1.5$)	3	82.91	84.08	3	82.91	84.08	+0.00
EMD	2	81.29	82.70	2	81.29	82.70	+0.00
SOFT ($\beta = 2$)	2	81.41	81.86	3	81.58	82.38	+0.52
SOFT ($\beta = 3$)	1	79.72	80.31	3	80.89	82.31	+2.00
SOFT ($\beta = 4$)	2	80.05	80.94	3	81.09	82.61	+1.67
SOFT ($\beta = 5$)	2	80.63	81.37	3	81.18	82.19	+0.82
CORAL	2	82.90	84.33	3	83.11	84.50	+0.17
MSE → Soft-QWK	2	83.24	84.31	2	83.24	84.31	+0.00
CORAL, head lr 10^{-3}	3	83.86	84.58	3	83.86	84.58	+0.00

The rule is worth up to +2.00 QWK points on the test set. It changes nothing for the four runs whose loss and QWK already peak at the same epoch, namely OLL, EMD, the two-phase method and the head learning rate ablation. It is worth +1.32 points for cross-entropy, +2.00 for SOFT at $\beta = 3$ and +1.67 for SOFT at $\beta = 4$, and only +0.17 for CORAL and +0.67 for the regression method.

Two conclusions follow. First, part of the difference between published systems that use different selection rules is the rule and not the model, and a comparison that does not hold the rule fixed is not a comparison of methods. Second, the effect is not uniform: it is largest for the methods whose loss is least aligned with the metric, and smallest for CORAL and regression, whose losses already reward proximity. The gaps between the methods in Table 4.5 shrink under QWK selection, because the methods that gain most are the ones that were behind.

4.10 The Two-Phase Soft-QWK Method

The two-phase method was trained as mean squared error for three epochs and then as Soft-QWK for three more, on the same six-epoch budget as every other run. The warm-up checkpoint, chosen by the same rule inside phase one, reaches 82.54 development QWK with 34.10 exact accuracy and a mean distance of 1.250. After the second phase the same model reaches 83.24 development QWK, that is +0.70 points, with adjacent accuracy rising to 72.33, the highest in the study, and the mean distance rising to 1.251, one of the two worst. On the test set the method reaches 84.31.

The size and the direction of that gain match what the STBW authors report for their own system, which improved from 83.9 to 84.88 QWK when the second phase was added [46]. Their absolute numbers are not comparable with ours, for two reasons: they evaluate on the shared-task splits, and their input carries corpus metadata and surface statistics in addition to the segmented text, while the input here is the segmented text alone.

Table 4.2 shows the phase-two trajectory, which rises from 82.13 to 83.24 and then falls to 82.57. A batch-level objective that can be lowered by predicting near the centre of the scale is expected to drift in exactly that way, and the metrics of the selected epoch show the drift beginning: adjacent accuracy is the best in the study while the mean distance and the exact accuracy are among the worst, which is the profile of a model that has learned to stay close to the middle. The warm-up is therefore not a formality, and the fact that this thesis reproduces the benefit of the second phase under a protocol that gives it no extra epochs is the substantive result here.

4.11 Loss Functions from Readability Research in Other Languages

This section reports the three losses of Section 3.7: CORN, and the two pairwise ranking objectives added to the regression loss, margin ranking (RANK) and pairwise logistic (PLR). They were trained after the main comparison, with the same script and protocol, and the same integrity checks passed in all of them: one accelerator, a realised batch of 64 sentences, 857 optimiser steps in every epoch, and a reloaded checkpoint that reproduces the logged development metrics exactly. Tables 4.8 and 4.9 give their results beside their matched controls, and Table 4.10 gives each difference to its control.

Table 4.8: Loss functions from readability research in other languages, on the BAREC development set, beside their matched controls from Table 4.4. Same pipeline, one head and one loss changed.

Method	QWK	Acc19	±1	Dist	Acc7	Acc5	Acc3	far≥3	far≥5	mF1	Ep.
Controls from the comparable group											
Regression (MSE)	82.88	35.54	71.15	1.212	57.02	64.43	73.34	13.43	3.67	22.13	2
CORAL	82.90	34.15	66.84	1.289	55.40	63.60	73.00	12.86	4.32	24.56	2
Loss functions added in this thesis											
CORN	81.92	54.47	71.46	1.096	65.39	70.05	76.79	13.83	5.34	42.57	3
MSE + margin ranking (RANK)	82.67	37.81	71.08	1.209	58.28	65.98	74.97	13.53	4.06	23.03	3
MSE + pairwise logistic (PLR)	82.61	30.70	69.77	1.269	55.50	63.52	73.89	14.30	3.34	18.21	2

QWK, the accuracies, far≥3, far≥5 and macro-F1 are percentages; Dist is in levels; Ep. is the selected epoch. far≥3 and far≥5 are this thesis's own metrics.

Table 4.9: Loss functions from readability research in other languages, on the BAREC test set, beside their matched controls from Table 4.5. Same pipeline, one head and one loss changed.

Method	QWK	Acc19	±1	Dist	Acc7	Acc5	Acc3	far≥3	far≥5	mF1	Ep.
Controls from the comparable group											
Regression (MSE)	83.64	34.20	71.70	1.186	56.24	63.05	71.31	13.41	2.88	23.61	2
CORAL	84.33	34.15	66.70	1.232	54.09	60.98	69.21	12.06	3.29	26.38	2
Loss functions added in this thesis											
CORN	82.57	54.39	71.49	1.058	64.69	69.02	75.20	13.34	4.87	43.97	3
MSE + margin ranking (RANK)	84.06	36.91	71.96	1.158	57.53	64.36	72.43	12.41	3.31	23.34	3
MSE + pairwise logistic (PLR)	83.68	30.19	70.74	1.219	56.03	63.19	71.89	13.08	2.54	18.63	2

QWK, the accuracies, far≥3, far≥5 and macro-F1 are percentages; Dist is in levels; Ep. is the selected epoch. far≥3 and far≥5 are this thesis's own metrics.

Table 4.10: Each added loss minus its matched control, in percentage points. A positive QWK or Acc19 difference is better; a negative far-miss difference is better. One seed per method.

		development				test			
Method	**Control**	QWK	Acc19	far≥3	far≥5	QWK	Acc19	far≥3	far≥5
CORN	CORAL	-0.98	+20.33	+0.97	+1.01	-1.76	+20.24	+1.28	+1.58
RANK	Regression (MSE)	-0.21	+2.27	+0.10	+0.40	+0.41	+2.70	-1.00	+0.43
PLR	Regression (MSE)	-0.28	-4.84	+0.86	-0.33	+0.04	-4.01	-0.33	-0.34

4.11.1 Margin Ranking

Adding the margin ranking term to the regression loss leaves QWK where it was: the difference to the regression run is -0.21 points on the development set and +0.41 on the test set. On the test set that places RANK, at 84.06, in the leading group of Section 4.4, beside ordinal log loss at 84.08; on the development set, at 82.67, it sits a few tenths below it. What the term changes is the error profile on the test set. Exact accuracy rises by +2.70 points and the share of errors of three levels or more falls by 1.00 points to 12.41 per cent, the lowest value outside the CORAL family. The hinge

only acts on pairs whose scores are in the wrong order or closer than the margin, which is where a regression that compresses the scale towards its centre makes its large errors. The development set does not show the same reduction, however (+0.10 points), so on one seed the effect is an indication rather than a measurement.

4.11.2 Pairwise Logistic

The pairwise logistic term also leaves QWK unchanged, at -0.28 points on development and +0.04 on test, but its effect on the errors is consistent across both splits: PLR has the fewest errors of five levels or more of every run in this thesis, 3.34 per cent on development and 2.54 per cent on test, below the regression method, which held that position before. It pays for this with exact accuracy, which falls by 4.01 points on the test set, and with the lowest macro-F1 of the table, 18.63. Unlike the hinge, the logistic loss never reaches zero, so it keeps pushing correctly ordered pairs further apart; the result is the most cautious model of the study, which rarely makes a large mistake and rarely commits to an exact level. The comparison between RANK and PLR, which differ only in the pair loss, shows that the choice between a hinge and a logistic function is not neutral: at the same QWK, one buys exact accuracy and the other buys the tail.

4.11.3 CORN

CORN is the loss the ZAI system paper names as future work [40], and its matched control is the CORAL head. It does not reach CORAL's QWK: it is 0.98 points lower on development and 1.76 lower on test. It changes the behaviour of the head completely, though. Its exact accuracy is 20.24 points higher than CORAL's on the test set, at 54.39 per cent, and its macro-F1 is +17.58 points higher, so it uses the whole scale; its mean distance on the test set, 1.058 levels, is the lowest of any run in this thesis; and its share of

errors of five levels or more is higher than CORAL's by +1.58 points. On the development set it has the highest QWK of all the methods whose exact accuracy is above 50 per cent, and on the test set it is second among them, behind EMD at 82.70.

This is the profile of the softmax heads of Section 4.5, not of CORAL, and it explains where that trade-off comes from. With a shared projection, CORAL computes one score per sentence and compares it with 18 thresholds, which is a regression with learned cut-points, and it behaves like one. Without the shared projection, each of CORN's 18 questions is a classifier of its own, and the head behaves like a classifier: it commits to levels, wins exact matches and occasionally commits far from the gold level. The shared projection, not the binary decomposition, is what gives CORAL its high QWK and its low exact accuracy.

4.11.4 Selection Rule and Summary

For CORN and RANK the epoch with the lowest development loss is also the epoch with the highest development QWK, so the selection rule of Section 4.9 changes nothing (+0.00 and +0.00 test QWK points). For PLR the QWK-selected epoch is one later and gives -0.04 points, so the rule does not favour any of the three.

In summary, the losses brought from readability research in other languages carry over to Arabic in the sense that matters here: the two ranking losses stay within a few tenths of a QWK point of their control, the margin ranking loss joins the leading group on the test set, and each of the three losses moves the error profile in a specific and interpretable direction. None of them beats the leading group on both splits, and with one seed per method the differences in QWK of a few tenths of a point are not measurements. The robust findings are the qualitative ones: the pairwise logistic loss is the most effective way found in this thesis to avoid errors of five levels or more, and removing CORAL's shared projection turns an ordinal head into a classifier.

4.12 Frozen-Backbone Track

Table 4.11 reports the second track, in which the encoder is frozen and only a linear head is trained on cached features, over five seeds.

Table 4.11: Frozen-backbone track: linear heads on cached AraBERTv2 features, mean ± standard deviation over five seeds, development set. These numbers are not comparable with the fine-tuning tables.

	pooler features				**mean-pooled features**			
Method	QWK	Acc19	Dist	far≥3	QWK	Acc19	Dist	far≥3
CE	71.52 ± 0.14	40.61	1.593	22.04	74.44 ± 0.00	46.34	1.447	20.32
Regression (MSE)	70.16 ± 0.13	19.76	1.723	24.32	75.57 ± 0.12	21.17	1.591	21.32
OLL ($\alpha = 1.5$)	63.39 ± 2.53	17.84	1.796	26.24	74.18 ± 0.04	29.27	1.504	21.41
EMD	72.84 ± 0.32	38.04	1.597	21.84	75.40 ± 0.36	43.77	1.445	19.78
SOFT ($\beta = 2$)	72.22 ± 0.13	40.55	1.578	21.87	75.09 ± 0.07	46.40	1.420	19.39
SOFT ($\beta = 3$)	71.93 ± 0.13	40.68	1.582	21.89	74.89 ± 0.08	46.43	1.429	19.76
SOFT ($\beta = 4$)	71.77 ± 0.15	40.67	1.586	21.92	74.68 ± 0.13	46.44	1.438	20.09
SOFT ($\beta = 5$)	71.60 ± 0.12	40.64	1.591	22.01	74.53 ± 0.14	46.40	1.443	20.22
CORAL	71.03 ± 0.17	25.14	1.687	23.34	75.53 ± 0.16	27.47	1.559	20.24
MSE → Soft-QWK	74.55 ± 0.06	17.90	1.838	26.85	78.00 ± 0.10	19.68	1.702	23.17
CORAL, zero-initialised	71.05 ± 0.18	25.15	1.686	23.33	75.50 ± 0.06	27.52	1.559	20.32

Three results deserve comment.

The first is the variance. Every method except one has a standard deviation over five seeds of at most 0.36 QWK points, so on frozen features the seed is nearly irrelevant. The exception is OLL on pooled features, at 2.53 points, which makes the point that the stability of a loss is itself a property worth reporting: the same loss that is among the leaders when the encoder can adapt is the least stable one when it cannot.

The second is the ranking. Under freezing, the metric-aligned two-phase method is the strongest on QWK on both representations, 74.55 and 78.00, and it remains the weakest on exact accuracy and mean distance, so the trade-off of Section 4.5 survives the change of regime. The ordering of the

remaining methods does not survive. The rank correlation between the fine-tuned ranking and the frozen ranking is -0.53 on pooled features and 0.25 on mean-pooled features, that is, no relationship at all in the second case and a mild inversion in the first. OLL, third when fine-tuned, is last under freezing on both representations, and cross-entropy climbs. A loss comparison on frozen features is therefore not a cheap substitute for a loss comparison under fine-tuning, and this thesis reports the frozen track for its variance and for its CPU-friendly setting rather than as evidence about the fine-tuning ranking.

The third is the cut-point ablation that the fine-tuning track could not afford. With the head trained for 50 epochs at 10^{-3}, the travel budget is 21.4, and the frequency initialisation of Equation 3.19 then makes no difference at all: the frequency-initialised and the zero-initialised heads differ by -0.01 QWK points on pooled features and +0.04 on mean-pooled features, which is inside the seed-to-seed spread. This is the other side of the argument in Section 4.7. When the budget is large the initialisation is irrelevant; when it is 0.13, as under the BAREC protocol, the cut-points never leave their initial values. The initialisation is a remedy for a specific arithmetic situation, not a general improvement, and the frozen track is what shows the difference.

4.13 Context: Published Systems and Human Agreement

Two comparisons frame these numbers, and both need a caution.

The published shared-task systems report QWK on the blind test set of the BAREC 2025 shared task: 85.5 for the CORAL system and 85.6 for the two-phase system [40, 46]. Those are different data from the corpus test split used here, and both systems differ from this study in more than the loss: the CORAL system selects on QWK, tunes its threshold and uses early stopping, and the two-phase system adds metadata features. The results here are therefore comparable with the benchmark tables, which they reproduce,

and not with the leaderboard.

Human agreement is the second frame. The corpus authors report an average pairwise QWK between annotators of 81.8 and an average QWK of 88.1 between individual annotators and the unified label, with an exact pairwise agreement of only 61.1 per cent and an average pairwise distance of 0.94 levels [24]. The best test QWK in this study, 84.58, is therefore above the agreement of two randomly chosen annotators and still below the agreement of an annotator with the adjudicated label. The exact-accuracy figures deserve the same framing: the best exact accuracy here, 56.03 per cent, is not far below what two annotators achieve with each other. This is the honest measure of how much room is left on this task, and it also explains why nineteen-way exact accuracy is a demanding criterion for a scale whose own annotators agree exactly in six cases out of ten.

4.14 Limitations

The following limitations bound every claim made above.

1. **One seed per method in the fine-tuning track.** The protocol fixes the seed at 42 and the accelerator budget did not allow repeats. Differences of a few tenths of a QWK point, which is the size of the gaps among the four leaders, cannot be separated from run-to-run variation. This is why no single best method is named. The frozen track shows that seed variation is small once the encoder is fixed, but it says nothing about variation in fine-tuning, where the encoder itself moves.

2. **Mixed precision.** The runs use fp16 autocast where the benchmark runs in single precision. The losses are computed in single precision, but the encoder forward and backward passes are not, and the number of optimiser steps skipped because of gradient overflow was not recorded. A small part of the differences in Table 4.1 may come from this.

3. **Batch composition.** The effective batch of 64 is assembled from four micro-batches of 16. This is arithmetically equivalent for per-sentence losses except in the last step of each epoch, as described in Section 3.9.

4. **The two-phase method is not in the comparable group.** It uses the same six epochs but a different schedule and, in its second phase, a true batch of 64 with gradient checkpointing. Its numbers answer whether the metric-aligned objective helps, not whether it would win a comparison in which every method had a two-phase schedule.

5. **The CORAL cut-point result is a measurement plus one ablation.** That the cut-points barely move is measured directly and is not in doubt. That the initialisation is therefore necessary is an argument supported by the movement, by the head learning rate ablation and by the frozen-track comparison; a zero-initialised CORAL run under the fine-tuning protocol was not performed and would be the direct test.

6. **The regression gap.** Section 4.3 attributes most of it to the selected epoch and supports that with the per-epoch values, but the reason our development loss for that method bottoms out two epochs earlier than the published run does remains unexplained.

7. **The frozen track is not comparable with the published tables** and uses head hyper-parameters chosen by us, as stated in Section 3.12.

8. **The added losses.** CORN, RANK and PLR were each trained once, after the main comparison, and two of their settings are ours: the ranking pairs are formed inside the training batch, because BAREC has no parallel versions of a text, and the weight of the ranking term is fixed at one without tuning. Their comparisons with the matched controls share the single-seed limitation above.

9. **Scope.** One encoder, one corpus, sentence level, and no ensembling,

no external data and no metadata features. Nothing here speaks to document-level readability or to other Arabic encoders.

4.15 Summary

The pipeline reproduces the published benchmark: seven of eight losses land within 1.5 QWK points of their published values and the eighth is explained by one checkpoint decision, which the per-epoch values in Table 4.2 make visible.

On the question the research brief asks, the answer has three parts. Ordinal-aware objectives beat the ordinal-blind baseline by up to three QWK points on development and four on test, and the CORAL head and the metric-aligned two-phase objective are both in the leading group. Within that group, made up of CORAL, the two-phase Soft-QWK schedule, ordinal log loss and plain regression, the differences are a few tenths of a point on one seed and no single winner can be declared. And the ordinal mechanisms buy their QWK specifically by removing far misses, not by reducing the average error: CORAL has the fewest errors of three levels or more on both splits, the regression method the fewest of five levels or more, while the lowest mean distance and the highest exact accuracy belong to a soft-label configuration two QWK points behind them.

Three further findings came out of the ablations. The cut-points of a CORAL head move less than 0.0703 under the BAREC learning rate, so they are effectively supplied by their initialisation rather than learned, and raising only the head learning rate moves them and improves both QWK and exact accuracy. The checkpoint selection rule is worth up to +2.00 test QWK points and is worth most to the methods whose loss is least aligned with the metric, so it has to be held fixed in any comparison. And the ranking of losses does not survive freezing the encoder, which means a cheap frozen-feature study cannot stand in for a fine-tuning study, even though it

is the setting in which seed variance can actually be measured.

The three losses added from readability research in other languages confirm this picture from a different direction. The pairwise margin ranking loss joins the leading group on the test set at 84.06; the pairwise logistic loss has the fewest errors of five levels or more of any run, 2.54 per cent on test; and CORN, which removes CORAL's shared projection, gives up 1.76 test QWK points for 20.24 points of exact accuracy and the lowest mean distance of the study, which shows that the shared projection is what makes CORAL behave like a regression.

The next chapter draws these findings together and states what remains to be done.

Chapter 5

Conclusion and Future Work

5.1 Summary of the Work

This thesis asked which ordinal head and which loss function serve fine-grained Arabic readability assessment best, and whether the errors they make are near misses or far misses. Before BAREC, Arabic readability work relied on formulas adapted from English, on classifiers trained on coarse school or learner scales, and on lexicons and corpora that label words or whole texts; BAREC made it possible, for the first time, to compare models on 19 expert-assigned levels for every sentence. The question came from a comparison that the literature leaves open: the BAREC benchmark [24] compares distance-aware losses but has no ordinal head and no metric-aligned objective, while the two shared-task systems closest to this work add a CORAL head [40] and a differentiable kappa objective [46] but each does so inside its own pipeline, with its own input, its own selection rule and its own evaluation data, so neither can be read as a measurement of what its ordinal mechanism contributes.

The answer was pursued by building one pipeline and changing one thing at a time. The corpus, the splits, the segmented input, the AraBERTv2 encoder and the whole training protocol were taken from the benchmark

script and held fixed, down to the library defaults that the script does not state; the prediction head and the loss function were the only components allowed to vary. Eleven runs were carried out on this basis: nine that differ only in head and loss, one two-phase metric-aligned run, and one ablation on the learning rate of the CORAL head. Three more runs then added losses that the benchmark does not evaluate, taken from readability research in other languages or named as future work by a BAREC system: CORN and two pairwise ranking objectives. Two more ablations re-evaluated saved models under a tuned decision threshold and under the alternative checkpoint selection rule, and a second track trained linear heads on frozen features over five seeds, which both satisfies the CPU-friendly requirement of the research brief and supplies the variance estimates that a single-seed protocol cannot.

Before any of it was interpreted, the implementation was tested against the sources rather than assumed to match them. Every loss was checked against an independent implementation to better than 10^{-9}, the metric code was checked against its definitions, the split sizes and the collapse mappings were checked against the corpus itself, and each run recorded the accelerator count, the realised batch, the observed number of optimiser steps and a comparison between the reloaded checkpoint and the training log. That record is what allows the numbers in Chapter 4 to be quoted at two decimal places.

5.2 Findings

5.2.1 The Pipeline Reproduces the Benchmark

Seven of the eight losses that the benchmark also reports land within 1.5 QWK points of their published development values, with a mean absolute difference of 0.64 points, and cross-entropy, EMD and ordinal log loss land within -0.19, +0.09 and +0.11 points. The eighth, the regression method, is

-1.12 points below its published value under the benchmark's own selection rule, and the per-epoch values show why: its development QWK reaches 83.55 two epochs after its development loss bottoms out, and at that epoch its test QWK is 84.32 against the published 84.4. The pipeline therefore reproduces the published work, and the one visible discrepancy is a property of the selection rule rather than of the implementation.

5.2.2 Ordinal Mechanisms Help, and They Are Equivalent among Themselves

Every ordinal-aware objective except one soft-label setting beats the ordinal-blind baseline, by up to three QWK points on the development set and four on the test set. The leading group consists of four objectives: the CORAL head (82.90 development, 84.33 test), the two-phase metric-aligned schedule (83.24 and 84.31), distance-weighted ordinal log loss (82.91 and 84.08) and plain regression (82.88 and 83.64). The differences inside that group are a few tenths of a QWK point, they do not keep the same order across the two splits, and each method was trained once, so this thesis declines to name a single winner. The defensible statement is that the ordinal head and the metric-aligned objective reach the level of the best distance-aware loss under an identical protocol, which is what the research brief asked, and that nothing in these results shows any of them to be better than the others.

5.2.3 The Metric Rewards Removing Far Misses, Not Reducing Average Error

Separating the error distances changed the picture that a single metric gives. The methods that lead on QWK are not the methods that predict most levels exactly, and they are not the methods with the smallest average error either. The lowest mean distance in the study, 1.088 levels, and the highest exact accuracy, 57.07 per cent, both belong to a soft-label configuration two QWK

points behind the leaders, whose own mean distances are the worst in the table. What the leaders do is control the tail: CORAL has the fewest errors of three levels or more on both splits (12.86 per cent on development, 12.06 on test) and the regression method the fewest of five levels or more (3.67 and 2.88), while cross-entropy is worst on both counts.

This is a usable result rather than a curiosity. A reader who needs a text placed within a narrow band should choose an ordinal head or a regression head; a reader who needs the exact level should choose a soft-label loss and accept two QWK points less agreement. Reporting QWK alone, as the benchmark and the shared task do, hides a difference of nearly twenty points of exact accuracy between methods whose QWK differs by two.

5.2.4 A CORAL Head Needs Its Cut-Points Initialised, Not Learned

Under the benchmark protocol, the 18 cut-points of the CORAL head never moved further than 0.0703 from their initial values, against a fitted spread of 11.68 between the highest and the lowest. The reason is arithmetic: six epochs of 857 steps at a learning rate of 5×10^{-5} decaying to zero sum to a travel budget of 0.13, so a cut-point starting at zero cannot reach a value near -6. Initialising the cut-points from the cumulative label frequencies of the training split, which is the one element of the method taken from neither the benchmark nor the two shared-task papers, removes the problem instead of spending the budget on it, and it keeps the cut-points correctly ordered throughout training.

Two further experiments support the diagnosis from both sides. Raising the head learning rate alone to 10^{-3}, which multiplies the budget to 2.57, lets the cut-points move by 1.59 and raises development QWK by +0.96 points and test QWK by +0.25; and on frozen features, where the head budget is 21.4, the initialisation makes a difference of at most +0.04 QWK points, that is, none. The initialisation is therefore a remedy for a specific arithmetic

situation, and a head-specific learning rate appears to be the better remedy where it is available.

5.2.5 The Checkpoint Selection Rule Has to Be Held Fixed

Selecting on development QWK rather than development loss is worth up to +2.00 test QWK points, and what it is worth is not uniform: it gives +2.00 points to a soft-label setting and +1.32 to cross-entropy, but only +0.17 to CORAL and nothing at all to ordinal log loss, EMD or the two-phase schedule. The rule helps exactly those methods whose loss is least aligned with the metric. Any comparison between published systems that use different rules is therefore partly a comparison of rules, and a like-for-like comparison has to fix the rule first. A related, smaller finding is that the tuned decision threshold which the CORAL system reports as a source of gain contributes +0.12 test QWK points here, once the cut-points carry the label frequencies.

5.2.6 Losses from Readability Research in Other Languages Carry Over

The three added losses behave on Arabic as their design predicts. The margin ranking loss of the multilingual Wikipedia model [47], added to the regression loss, reaches 84.06 test QWK, inside the leading group, and lowers the share of errors of three levels or more on the test set by 1.00 points. The pairwise logistic loss of the neural pairwise ranking model [31] keeps QWK at the level of regression and has the fewest errors of five levels or more of any run on both splits (3.34 and 2.54 per cent). CORN [44], the objective the ZAI system names as future work, gives up 1.76 test QWK points against CORAL but gains 20.24 points of exact accuracy and the lowest mean distance of the study. That last result also explains the trade-off of the main comparison: it

is CORAL's shared projection, which turns the head into a regression with thresholds, that gives it its high QWK and its low exact accuracy.

5.2.7 A Frozen-Feature Study Cannot Stand In for a Fine-Tuning Study

On frozen features the seed hardly matters, with a standard deviation over five seeds of at most 0.36 QWK points for every method but one, and the exception, ordinal log loss at 2.53 points, is itself informative about that loss. But the ranking of the losses does not survive freezing: the rank correlation with the fine-tuned ranking is -0.53 on pooled features and 0.25 on mean-pooled features. The cheap protocol answers questions about variance and about CPU-friendly training; it does not answer the question about which loss to use when the encoder can adapt.

5.3 Contributions

1. A controlled comparison of nine head and loss configurations for 19-level Arabic readability on one fixed pipeline, extending the benchmark's four losses with the CORAL head, the four soft-label settings and a metric-aligned two-phase objective, and reporting them in one table on both official splits.

2. The first evaluation on BAREC of three losses from outside the benchmark: CORN and the pairwise margin ranking and pairwise logistic losses of readability research in other languages, each against a matched control, with the finding that the pairwise logistic loss minimises the errors of five levels or more and that CORN trades QWK for exact accuracy.

3. A review of Arabic readability work before BAREC that states, for

each formula, classifier, lexicon and corpus, what it could not support for fine-grained sentence-level modelling.

4. A verified reproduction of the benchmark's loss comparison, with the one visible discrepancy traced to the checkpoint selection rule.

5. An initialisation of the CORAL cut-points from training label frequencies, with a quantitative argument from the travel budget of the schedule, a direct measurement of how far the cut-points actually move, and two ablations that test the argument from opposite directions.

6. Two metrics of error distance, the share of errors of at least three and at least five levels, which make the near-miss and far-miss behaviour of each loss visible, and the finding that QWK leadership is bought by removing far misses rather than by reducing average error.

7. A quantification of how much the checkpoint selection rule and the CORAL decision threshold are worth, which is what makes published systems with different protocols comparable.

8. A frozen-backbone track that satisfies the CPU-friendly requirement of the research brief, supplies five-seed variance estimates, and shows that the loss ranking does not transfer between the two regimes.

9. A reproducibility record: verified losses and metrics, asserted data, recorded batch composition and optimiser steps, pinned corpus and encoder revisions, and a checkpoint check on every reported number.

5.4 Limitations

Section 4.14 states the limitations in full. The three that bound the conclusions most tightly are: one seed per method in the fine-tuning track, so that differences of a few tenths of a QWK point are not measurements; mixed

precision, where the benchmark uses single precision; and the absence of a zero-initialised CORAL run under the fine-tuning protocol, which would be the direct test of the cut-point argument that the measurements and the two ablations only support indirectly.

5.5 Future Work

1. **Repeat the comparable group over several seeds.** This is the single most valuable next step, and the cheapest to justify: at eleven runs of roughly half an hour each on one mid-range accelerator, five seeds for the nine comparable methods is a day of computation and would turn the leading group from an observation into a measurement.

2. **Run CORAL with zero-initialised cut-points under the fine-tuning protocol.** One run would convert the cut-point argument from a supported diagnosis into a direct result.

3. **Separate the head learning rate from the encoder learning rate for every method.** The ablation here suggests that the protocol's single learning rate, chosen for an encoder, underserves any head with parameters that must reach specific values. Ordinal log loss and the soft-label family may be affected in the same way as CORAL.

4. **Tune and repeat the added losses.** The weight of the ranking term was fixed at one and the margin taken from the source paper; a small sweep over both, over several seeds, would show whether the reduction in far misses seen with the ranking losses is robust. Combining the pairwise logistic term with an ordinal head such as CORAL is a natural next objective.

5. **Give every method a two-phase schedule.** The metric-aligned objective was tested as its authors define it, which leaves open whether

a second phase of Soft-QWK would also improve CORAL or ordinal log loss. The comparison would then be between objectives under identical schedules.

6. **Report the error distance distribution as standard.** The far-miss metrics introduced here cost nothing to compute and expose behaviour that QWK and mean distance both hide; a per-level error analysis, of the kind the benchmark reports for its best model, would extend this to the question of *where* on the 19-level scale each loss fails.

7. **Extend beyond one encoder and one granularity.** The comparison is confined to AraBERTv2 at sentence level. Whether the same ordinal mechanisms behave the same way on other Arabic encoders, and at document level, is untested here.

5.6 Closing Remarks

Fine-grained Arabic readability assessment is a task on which two expert annotators agree exactly six times in ten and reach a pairwise agreement of 81.8 QWK [24]. Against that background, the best test agreement obtained in this work, 84.58, sits above the agreement of two annotators with each other and below the agreement of an annotator with the adjudicated label, which is where the remaining room for improvement lies.

The practical lesson of the thesis is narrower than a leaderboard result and more useful for building a system. Choosing an ordinal mechanism does matter: it is worth several QWK points over an ordinal-blind baseline, and it buys those points by not making large mistakes. But among the ordinal mechanisms that are properly implemented and properly initialised, the evidence here does not separate them, while two things that are usually treated as implementation details, the checkpoint selection rule and the learning rate given to the head, are worth as much as the choice of loss. A comparison that

does not hold those fixed is measuring the protocol rather than the method.

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